

Interacting Systems and how to treat them in Materials Theory

2026-09-11

(Final Seminar)

Sebastian Kalhöfer



It was the best of times, it was
the worst of times.

~ Charles Dickens

Projects (*those* that eventually *did* work)

- Green-functions
- Electron Spin Resonance
- Heat Capacities
- Fermi-Surfaces
- Weyl-Points

1 / 5 : Green-Functions

1 / 5 : Green-Functions

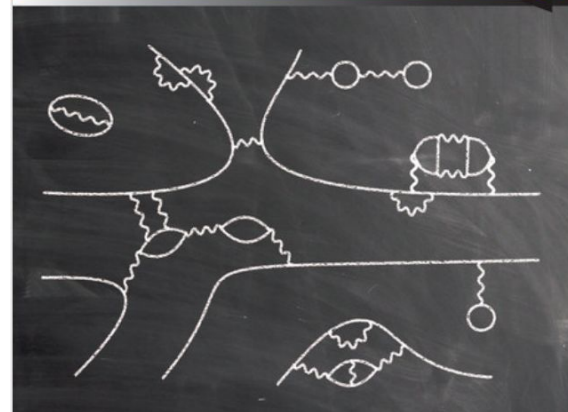


“The Green’s function method is one of the most powerful and versatile formalisms in physics”

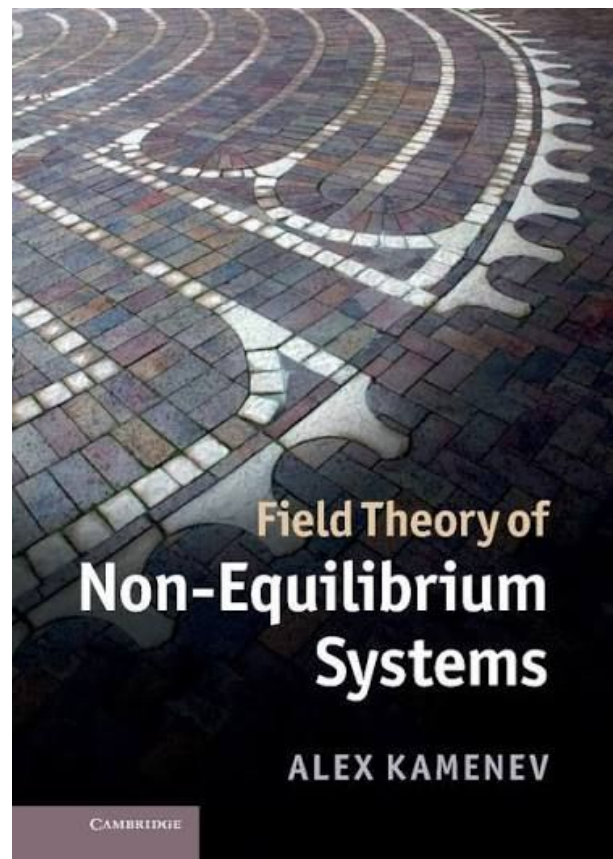
Gianluca Stefanucci
Robert van Leeuwen

Nonequilibrium Many-Body Theory of Quantum Systems

A Modern Introduction



"I can see how one can be overwhelmed by a number of different Green-functions, rules to follow and by the length of the calculations."





European Journal of Physics

REVIEW • OPEN ACCESS

Quantum many-body Green-functions: theory, interrelations and applications

Sebastian Kalhöfer

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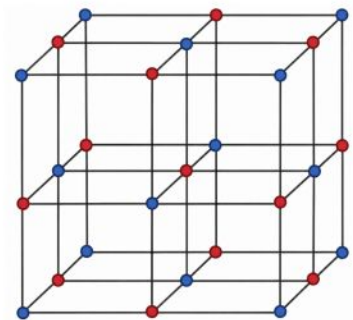
DOI 10.1088/1361-6404/ae26fc



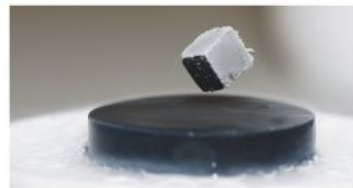
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System ("Cartoon"):

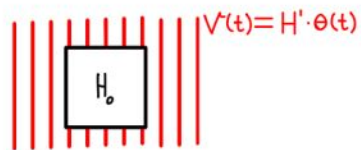
- Any arbitrary Crystal:



- Superconductor⁷:



- An *arbitrary* observable A of a System subjected to a „weak“ Perturbation H' :



Observable(s):

- Electronic Density of States / Band-Structure:

$$\rho_{nk}(\omega) = -\frac{\hbar}{\pi} \lim_{\epsilon \rightarrow 0} \text{Im} \lim_{\mathbf{k}, \mathbf{k}} G_{r,n}^{\text{ret}}(\omega + i\epsilon)$$

- Magnetization⁵:

$$\langle M \rangle \propto \sum_{\substack{\mathbf{k} \in \mathcal{Q} \\ n \in \{1, 2, \dots\} \\ \omega \in (-\infty, +\infty)}} G_{n,n}^<(\omega) - G_{n,n}^<(\omega)_{\substack{\mathbf{k}, \mathbf{k} \\ \uparrow, \uparrow \\ \downarrow, \downarrow}}$$

- Electronic / Phononic / Total Heat Capacity:

$$C_{N,V}(T) = \partial_T \langle H \rangle_{N,T} = \text{"sum of Green-functions"}^6$$

- Electronic Conductivity Tensor:

$$\sigma(\omega) = \text{"Green-function with current operators"}$$

- Thermal Conductivity Tensor:

$$\kappa = \text{"Green-function with energy-current operators"}$$

- Order Parameter⁸ (in Eliashberg-Theory):

$$\Delta_n(k) \propto \hat{\mathcal{G}}_{\substack{n \\ 1,2 \\ \text{Gorkov}}}(k)$$

- Kubo-Formula:

$$\langle A \rangle(t) = \langle A \rangle(t = -\infty) + \int dt' G_{AH'}^{\text{ret}}(t, t') + \mathcal{O}(H')^2$$

Equilibrium Statistical Physics

Quantum Many Body Physics

Equilibrium Statistical Physics



- Density Matrices
von Neumann Equation
- Fundamental Postulate of Statistical Physics

Microcanonical Ensemble

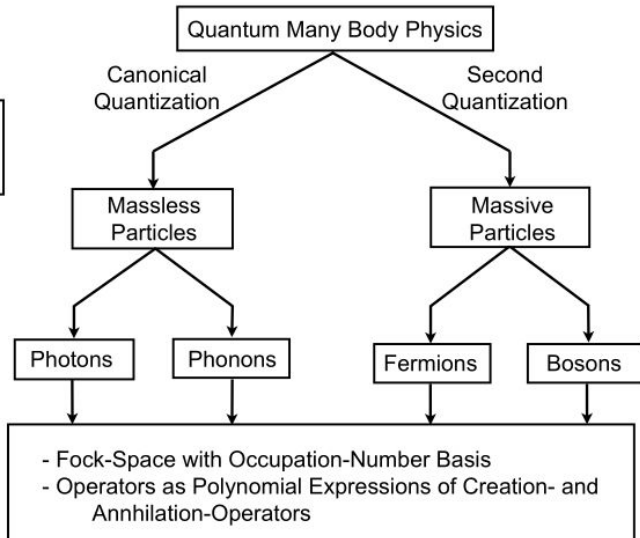
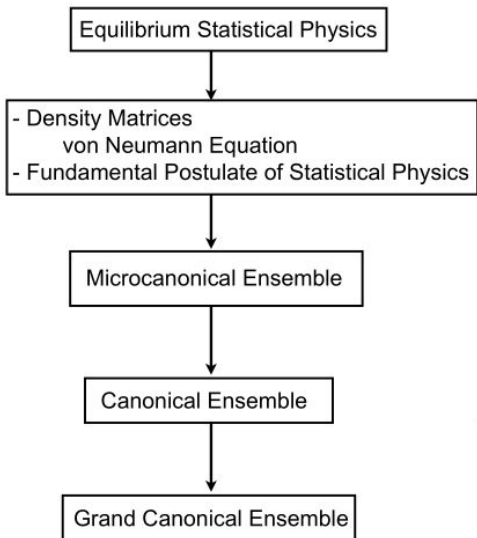


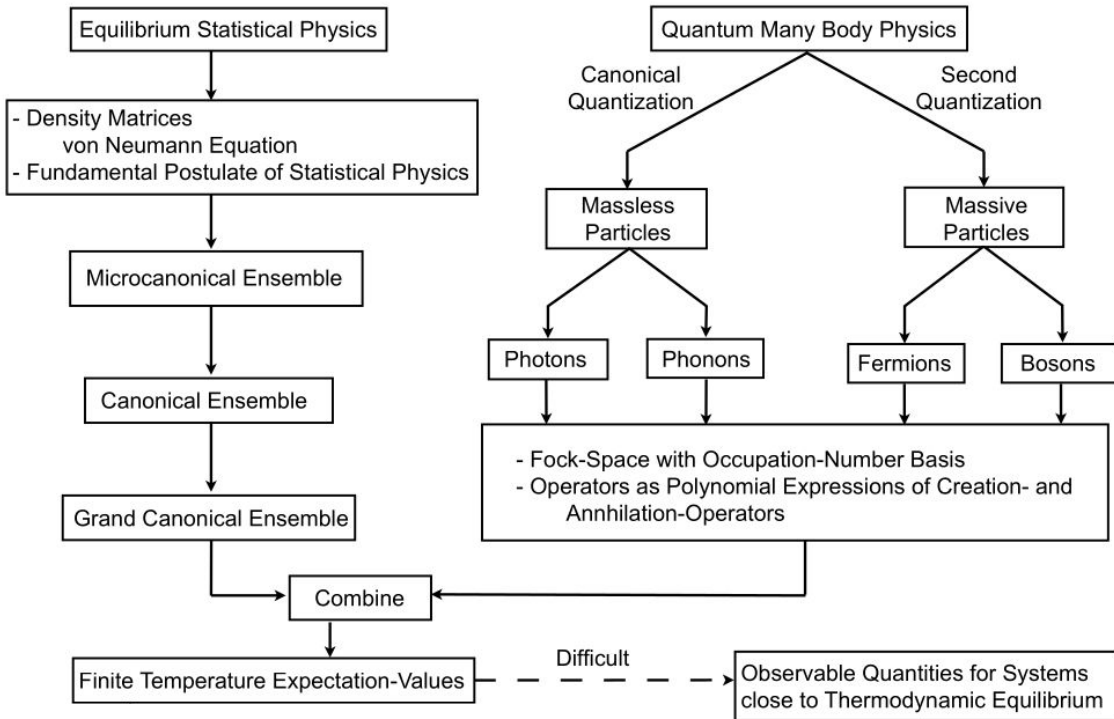
Canonical Ensemble

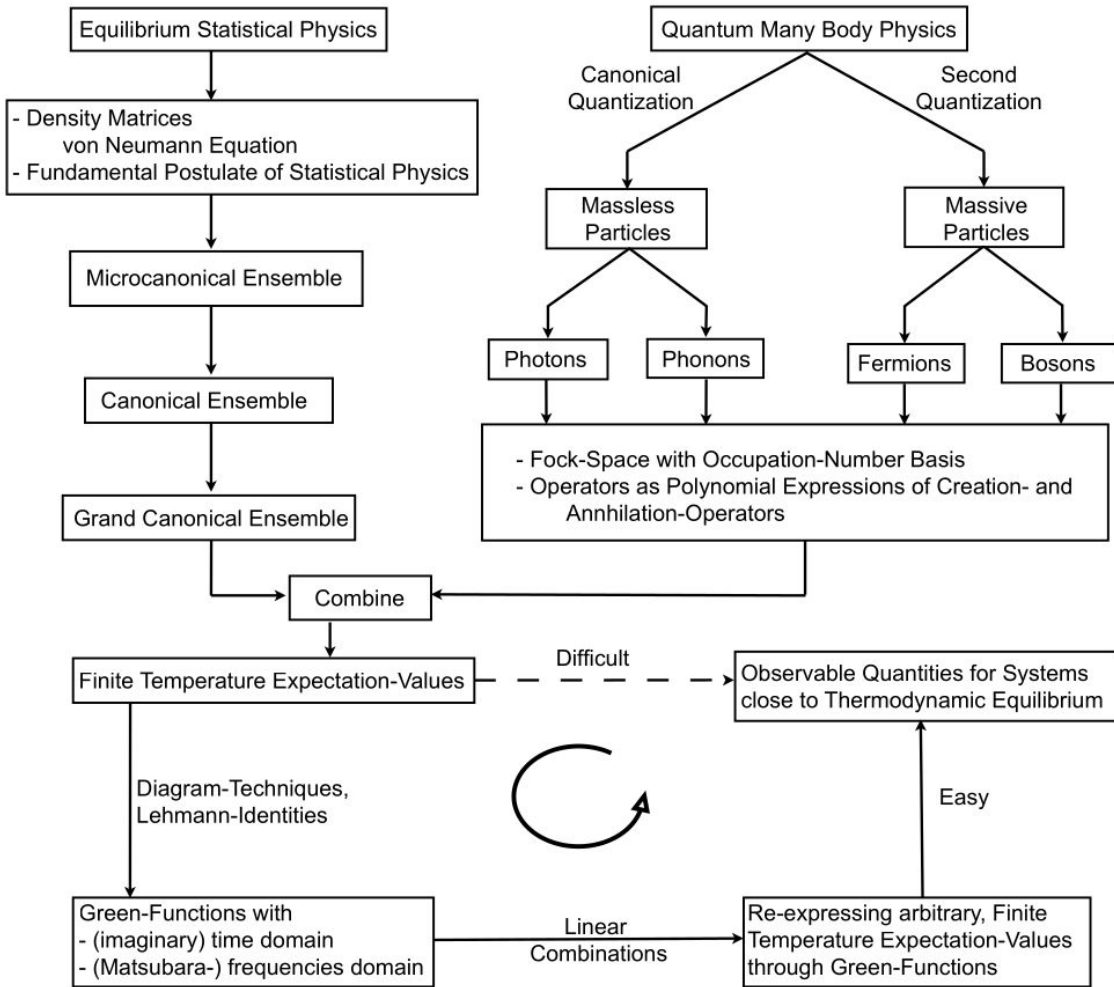


Grand Canonical Ensemble

Quantum Many Body Physics







Definition 7.26 We introduce the *statistical occupation-numbers* as set

$$\mathbb{N}_\zeta = \begin{cases} \{0, 1\} & \text{for } \zeta = -1, \\ \{0, 1, 2, \dots, \infty\} & \text{for } \zeta = +1. \end{cases}$$

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Notation 8.1 When we speak about "Green-functions" (to be introduced later), we consider one of the following Hilbert-spaces:

$$(1) \quad \mathcal{H} \equiv \mathcal{H}_{m=0} = \text{span}_{\mathbb{C}} \left\{ \prod_{j=1}^d \frac{1}{\sqrt{n_j!}} \left(c_j^\dagger \right)^{n_j} | \text{vac.} \rangle \equiv | n_1, n_2, \dots, n_d \rangle \left| n_1, n_2, \dots, n_d \in \mathbb{N}_\zeta \right. \right\} \quad (\text{massless particles}),$$

$$(2) \quad \mathcal{H} \equiv \mathcal{H}_{m>0} = \text{span}_{\mathbb{C}} \left\{ \prod_{j=1}^{d_H} \frac{1}{\sqrt{n_j!}} \left(c_j^\dagger \right)^{n_j} | \text{vac.} \rangle \equiv | n_1, n_2, \dots, n_{d_H} \rangle \left| n_1, n_2, \dots, n_{d_H} \in \mathbb{N}_\zeta \right. \right\} \quad (\text{massive particles}),$$

$$(3) \quad \mathcal{H} = \mathcal{H}_{m>0} \otimes \mathcal{H}_{m=0} \qquad \qquad \qquad (\text{massive-massless-mixtures}),$$

$$(4) \quad \mathcal{H} = \mathcal{H}_{m_1>0} \otimes \mathcal{H}_{m_2=0} \otimes \mathcal{H}_{m_3=0} \qquad \qquad \qquad (\text{massive-massless-massless-mixtures}),$$

$$(5) \quad \mathcal{H} = \mathcal{H}_{m_1>0} \otimes \mathcal{H}_{m_2>0} \otimes \mathcal{H}_{m_3=0} \qquad \qquad \qquad (\text{massive-massive-massless-mixtures}),$$

⋮

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- $\vdots$

Definition 8.3 We introduce the so-called *ensemble Hamiltonian* Ξ as

$$\Xi = \begin{cases} H & \text{for } \eta = 0 \text{ (massless particles),} \\ H - \mu N & \text{for } \eta = 1 \text{ (massive particles),} \\ H_{m>0} - \mu N_{m>0} + H_{m=0} + V_{\text{int}} & \text{for massive-massless-mixtures,} \\ \vdots & \end{cases} \tag{8.2}$$

12.2. Pure One-Body Green-Functions without Periodicity

Definition 12.5 We consider $X = c_\alpha$ and $X' = c_\alpha^\dagger$, for multi-indices $A \equiv (\alpha, t) \in \mathcal{A} \times \mathbb{R}$, with

12.2. Pure One-Body Green-Functions without Periodicity

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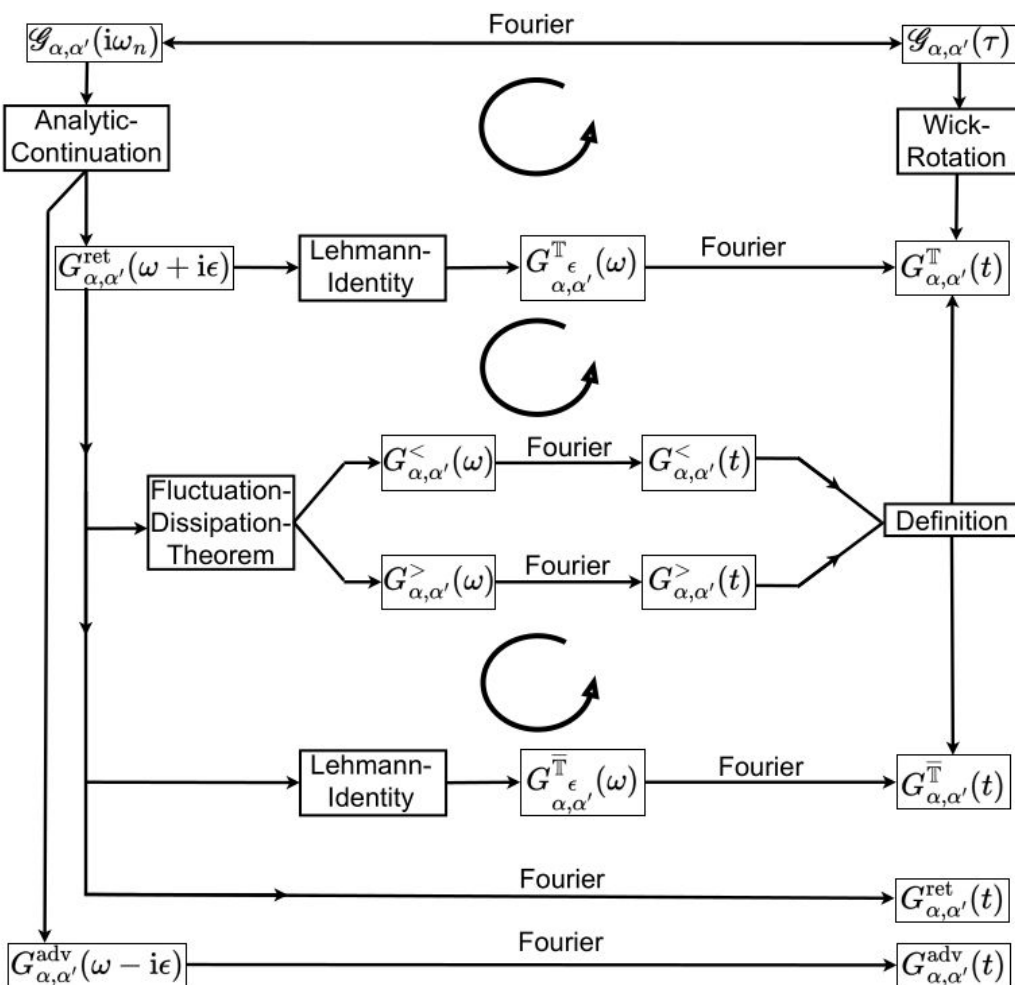
$$[c_\alpha, c_{\alpha'}^\dagger]_\zeta = \delta_{\alpha, \alpha'}, \quad S(c_\alpha) = S(c_{\alpha'}^\dagger) = \zeta.$$

Then we write for the *real-time-one-body-Green-function*², the *lesser*, *greater*, *anti-time-ordered*, *retarded* and *advanced Green's-functions* in their various shapes as *expectation-value*

- $G(A, A') \equiv G^{++}(A, A') \equiv G^\top(A, A') \equiv -\frac{i}{\hbar} \left\langle T_t c_\alpha(t) c_{\alpha'}^\dagger(t') \right\rangle,$
- $G^{+-}(A, A') \equiv G^<(A, A') \equiv -\frac{i}{\hbar} \zeta \left\langle c_{\alpha'}^\dagger(t') c_\alpha(t) \right\rangle,$
- $G^{-+}(A, A') \equiv G^>(A, A') \equiv -\frac{i}{\hbar} \left\langle c_\alpha(t) c_{\alpha'}^\dagger(t') \right\rangle,$
- $G^{--}(A, A') \equiv G^{\bar{\top}}(A, A') \equiv -\frac{i}{\hbar} \left\langle \bar{T}_t c_\alpha(t) c_{\alpha'}^\dagger(t') \right\rangle,$
- $G^{\text{ret}}(A, A') = -\frac{i}{\hbar} \theta(t - t') \left\langle [c_\alpha(t), c_{\alpha'}^\dagger(t')]_\zeta \right\rangle,$
- $G^{\text{adv}}(A, A') = +\frac{i}{\hbar} \theta(t' - t) \left\langle [c_\alpha(t), c_{\alpha'}^\dagger(t')]_\zeta \right\rangle.$

with time-ordering-operator and ensemble-Heisenberg-time-evolution as usual.

Green's Chart



holds for

- **non-interacting systems**

AND for

- **interacting systems!**

Lemma 14.13 For a given chemical potential μ , we can compute the average number of particles as

$$\langle N \rangle = \frac{-1}{\pi} \lim_{\epsilon \rightarrow 0} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} n_{\beta}^{\zeta}(E) \operatorname{Im} G_{\alpha, \alpha}^{\operatorname{ret}}(E + i\epsilon)$$

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Proof 26.45 of Lemma 14.13

We re-express the average particle number as lesser Green-functions and then use the fluctuation-dissipation-theorem

$$\begin{aligned} \langle N \rangle &= \sum_{\alpha \in \mathcal{A}} \langle c_{\alpha}^{\dagger} c_{\alpha} \rangle \\ &= -\zeta \frac{\hbar}{i} \sum_{\alpha \in \mathcal{A}} G_{\alpha, \alpha}^{<}(t = 0) \\ &= -\zeta \frac{1}{2\pi i} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} G_{\alpha, \alpha}^{<}(E) \\ &= \frac{-1}{\pi} \lim_{\eta \rightarrow 0} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} n_{\beta}^{\zeta}(E) \operatorname{Im} G_{\alpha, \alpha}^{\text{ret}}(E + i\eta) \end{aligned} \tag{26.77}$$

2 / 5 : Electron Spin Resonance

OPEN ACCESS

Temperature and magnetic field dependent g -factors in electron spin resonance spectroscopy

[Sebastian Kalhöfer](#) *

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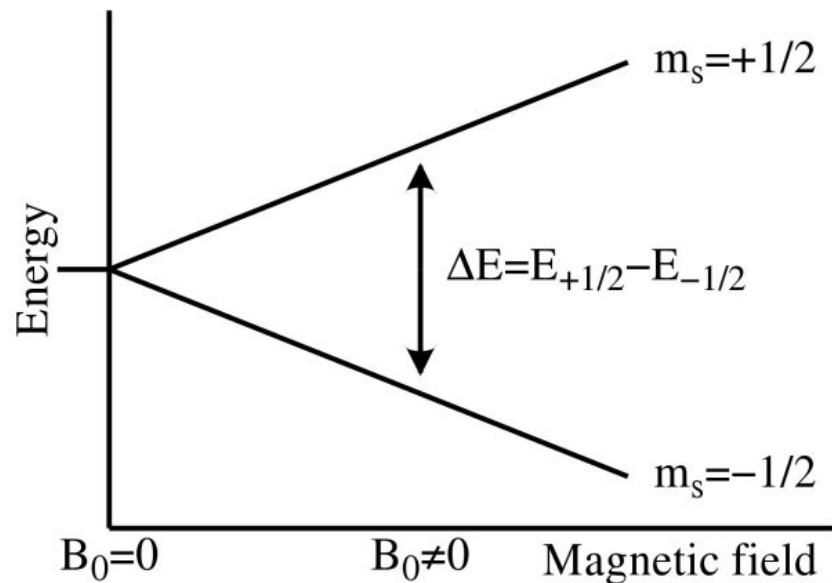
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Electron-Spin-Resonance:

Notation 1.1 Consider a single particle¹ with charge q_e subject to the Hamiltonian²

$$H_0 \equiv -\boldsymbol{\mu} \cdot \mathbf{B} \equiv -\gamma_e \mathbf{S} \cdot \mathbf{B} \equiv g_e \frac{-q_e}{2m_e} \mathbf{S} \cdot \mathbf{B} \equiv g_e \frac{-q_e}{2m_e} \frac{\hbar}{2} \boldsymbol{\sigma} \cdot \mathbf{B} \equiv g_e \frac{|e|}{2m_e} \frac{\hbar}{2} \boldsymbol{\sigma} \cdot \mathbf{B}$$

with single-particle Hilbert-space $\mathcal{H} \equiv \mathbb{C}^2$.





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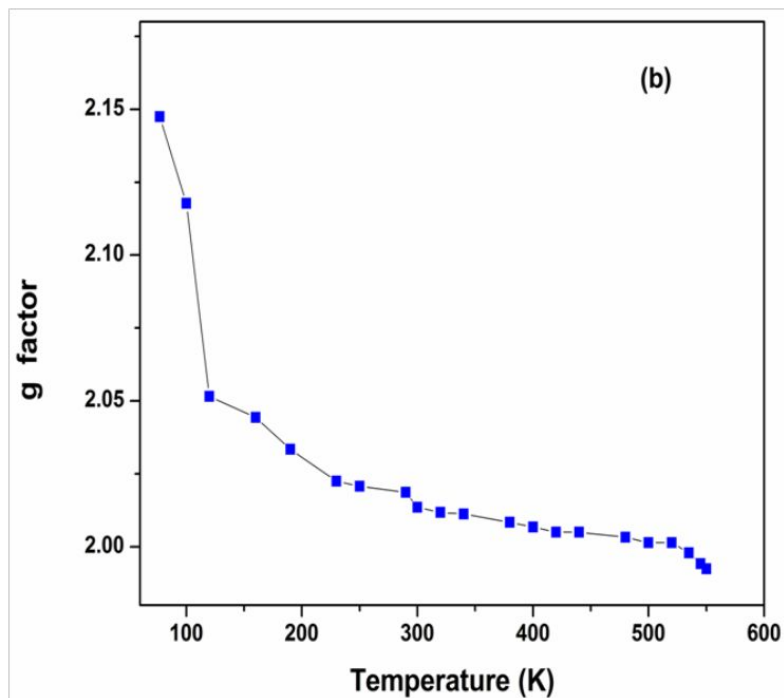
Journal of Magnetism and Magnetic Materials

journal homepage: www.elsevier.com/locate/jmmm



Temperature dependent electron paramagnetic resonance (EPR) of SrZrO_3

Santosh K. Gupta^{a,*}, Nimai Pathak^a, P.S. Ghosh^b, B. Rajeshwari^{a,b}, V. Natarajan^b,
R.M. Kadam^a





Temperature dependence of electron spin resonance in $\text{CaCu}_3\text{Ti}_4\text{O}_{12}$ substituted with transition metal elements

Virginie Brizé^{a,*}, Cécile Autret-Lambert^a, Jérôme Wolfman^a, Monique Gervais^a, Patrick Simon^{b,c}, François Gervais^a

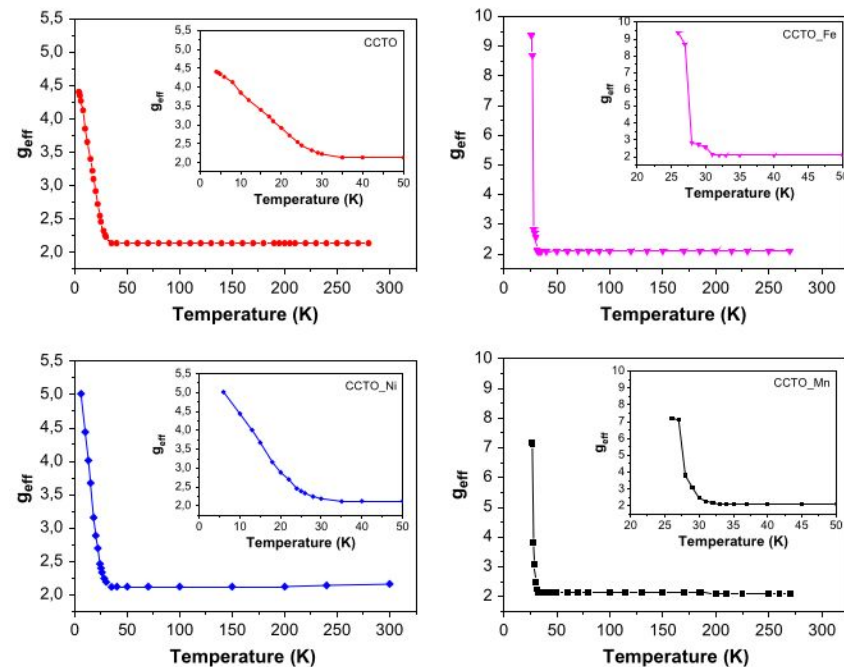


Fig. 6. Temperature dependence of g_{eff} of the samples of Fig. 3.

Temperature dependence of phase separation and magnetic anisotropy by electron spin resonance in $\text{Pr}_{0.6}\text{Ca}_{0.4}\text{Mn}_{0.9}\text{Ru}_{0.1}\text{O}_3$

C. Autret-Lambert^{1,a}, M. Gervais¹, M. Zaghrioui¹, S. Roger¹, F. Gervais¹, N. Raimboux², and P. Simon²

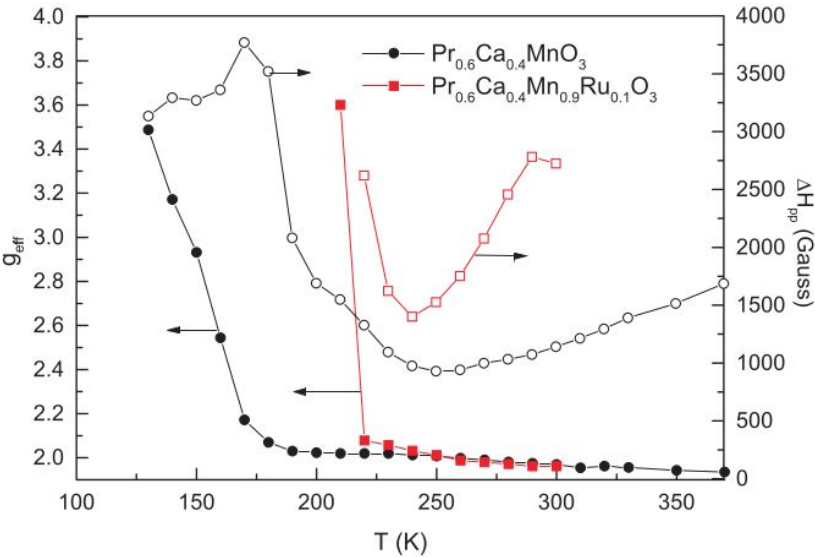


Fig. 5. Temperature dependence of the lineshape parameters for $\text{Pr}_{0.6}\text{Ca}_{0.4}\text{MnO}_3$ and $\text{Pr}_{0.6}\text{Ca}_{0.4}\text{Mn}_{0.9}\text{Ru}_{0.1}\text{O}_3$, g_{eff} -factor (left y -axis), linewidth (right y -axis).

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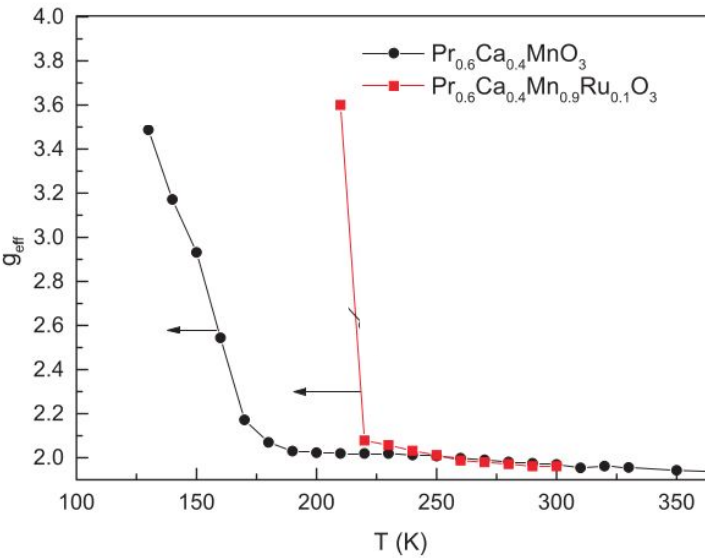


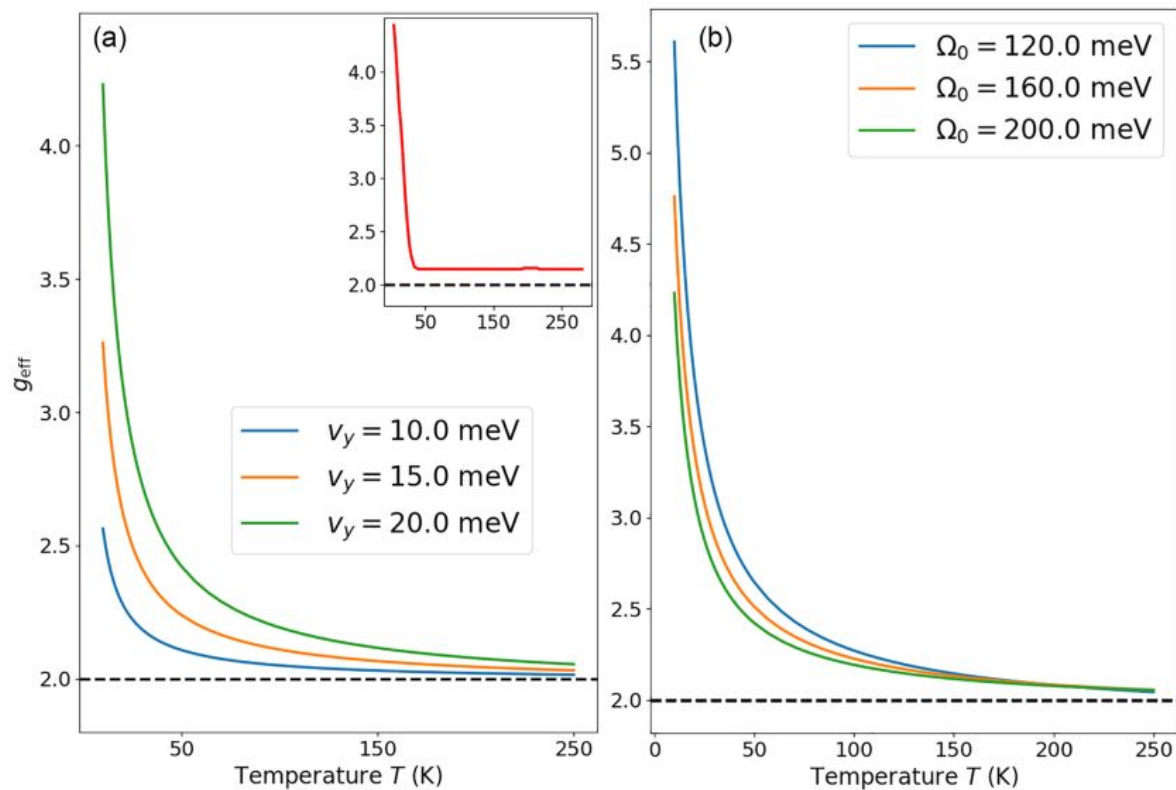
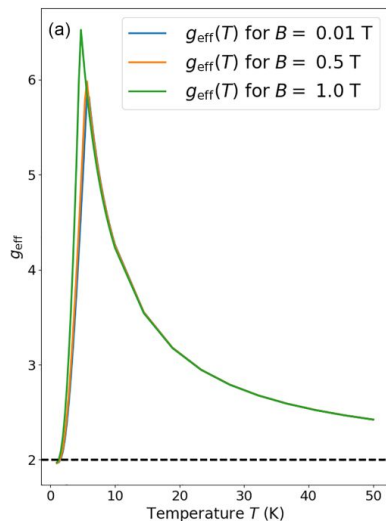
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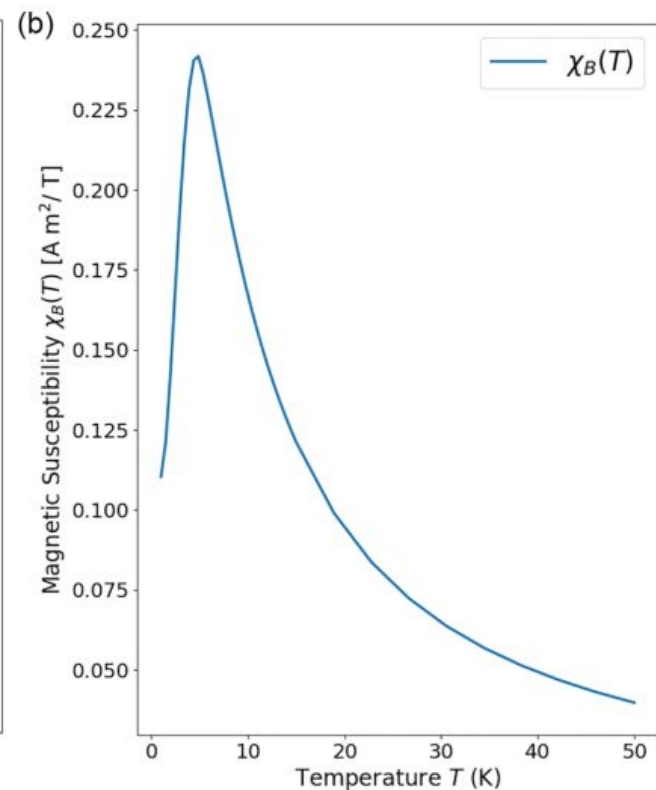
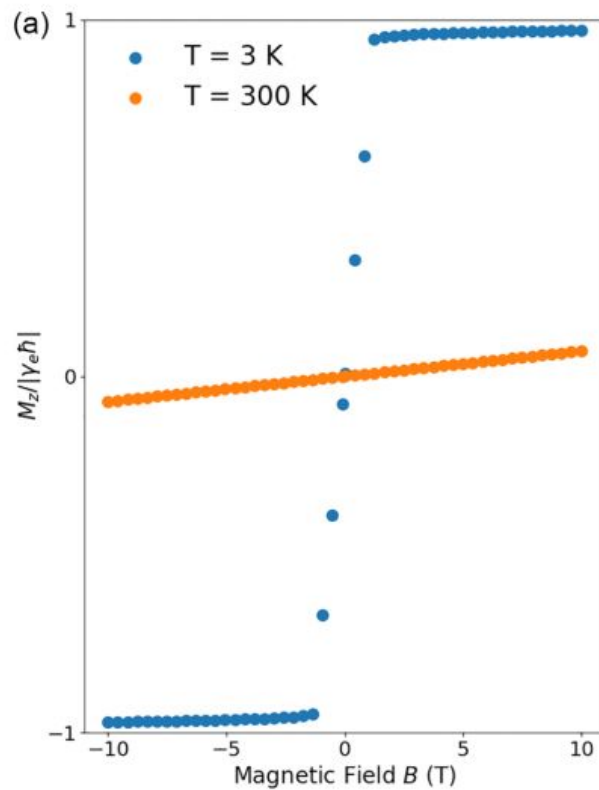
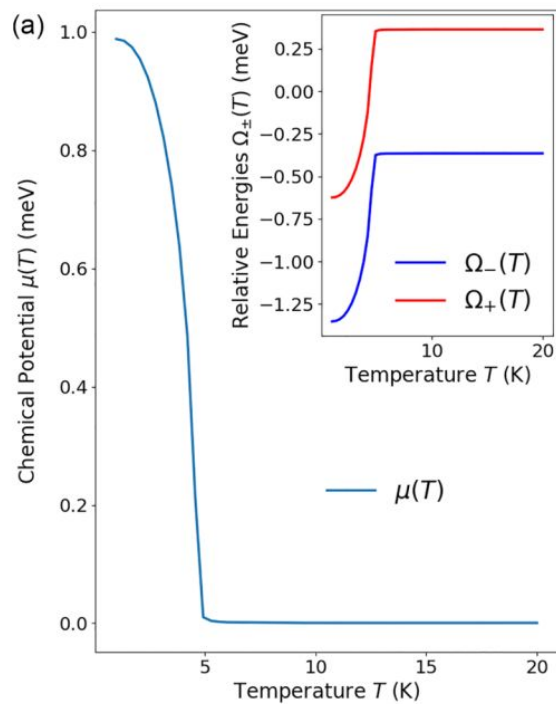
$$H_{\text{tot}} = H_{\text{el}} + H_{\text{ph}} + H_{\text{int}}$$

$$H_{\text{el}} = \psi^\dagger \epsilon \psi$$

$$H_{\text{ph}} = \hbar\omega b^\dagger b = \Omega b^\dagger b$$

$$H_{\text{int}} = \sum_{mn} \psi_m^\dagger V_{mn} \psi_n (b + b^\dagger)$$







UPPSALA
UNIVERSITET

Ph.D. position in Quantum Matter Theory

Project description: We are looking for a highly motivated and ambitious Ph.D. student in quantum condensed matter theory for a studies of magnetism and collective phenomena. First and foremost the studies concern effects of ionic vibrations on, for instance, magnetic ordering, but also other ordered structures, for example, superconductivity, are of interest. The studies will be conducted through both analytical and numerical methods, where the student will be given large freedom to define suitable methods.

3 / 5 : Heat Capacities

[Currently still a “draft”]

First-Principles Finite-Temperature Heat Capacities of Interacting Electron-Phonon Systems from a Functional Integral Approach

Sebastian Kalhöfer^{*} and Oscar Grånäs

Department of Physics and Astronomy, Uppsala University, Box 516, 75120 Uppsala, Sweden

Definition 9.31 We define the *heat capacity* of a system in the canonical ensemble as derivative

$$C_{V,N}(T) = T \left(\frac{\partial S}{\partial T} \right)_{V,N} (T) \quad (9.44)$$

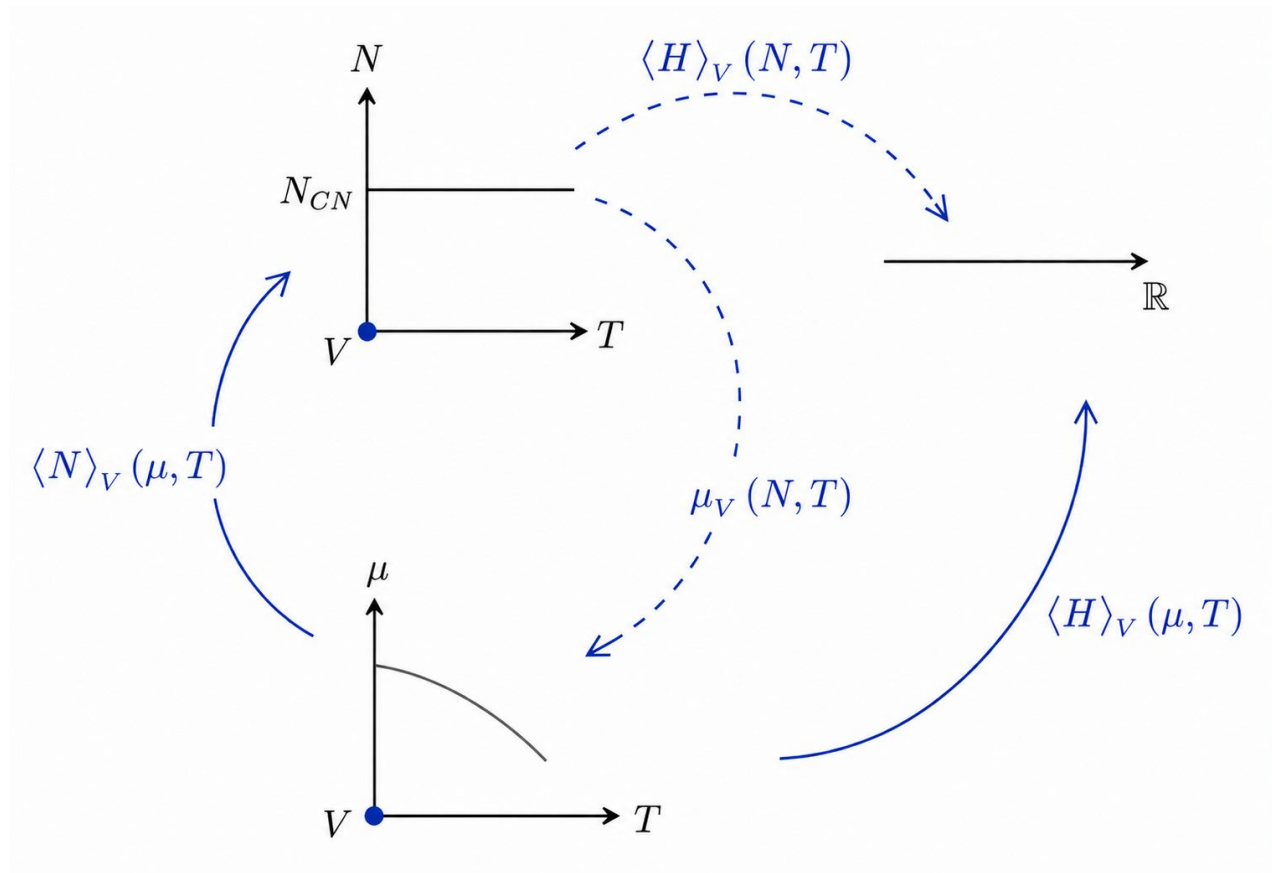
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Lemma 9.32 The following identities are immediate

- $C_{V,N}(T) = \partial_T \langle H \rangle_{V,N}(T)$
- $C_{V,N}(T) = -T \left(\partial_T^2 F \right) (N, T, V)$
- $C_{V,N}(T) = k_B \beta^2 \left(\langle H^2 \rangle_{N,V}(T) - \langle H \rangle_{N,V}^2(T) \right) = k_B \beta^2 \langle (H - \langle H \rangle)^2 \rangle_{N,V,T} \geq 0$

(9.45)



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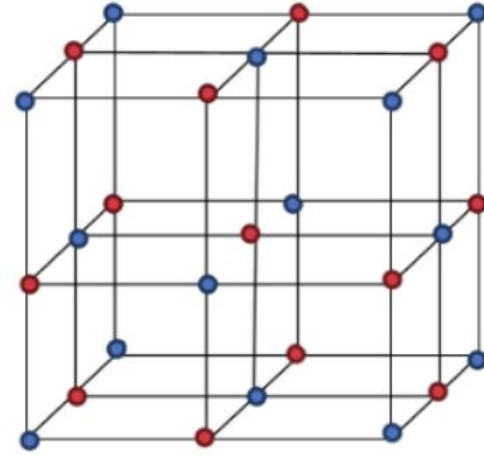
$$\begin{aligned} \bullet \quad & C_{V,N}(T) = \partial_T \langle H \rangle_{V,N}(T) \\ \bullet \quad & C_{V,N}(T) = -T \left(\partial_T^2 F \right) (N, T, V) \\ \bullet \quad & C_{V,N}(T) = k_B \beta^2 \left(\langle H^2 \rangle_{N,V}(T) - \langle H \rangle_{N,V}^2(T) \right) = k_B \beta^2 \langle (H - \langle H \rangle)^2 \rangle_{N,V,T} \geq 0 \end{aligned} \quad (9.45)$$

Corollary 9.38 Computing the specific heat capacity in the grand canonical ensemble one finds

$$\begin{aligned} C_{N,V}(T) &= \partial_T \langle H \rangle (N, T, V) \\ &= \left(\partial_T \langle H \rangle \right) (\mu(T, V), T, V) + \left(\partial_\mu \langle H \rangle \right) (\mu(T, V), T, V) \cdot \left(\partial_T \mu \right) (N, V, T) \end{aligned} \quad (9.53)$$

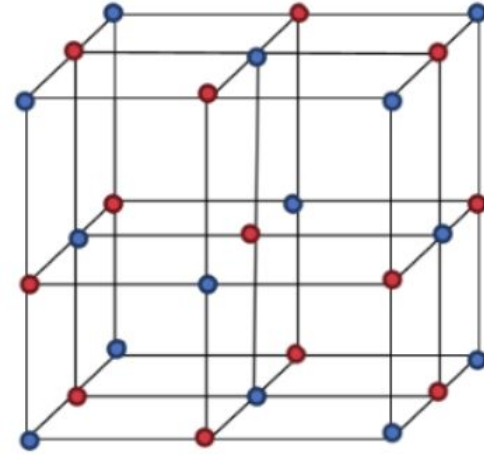
whereas the particle number N is kept constant.

Quick Repetition on Material-Theory:



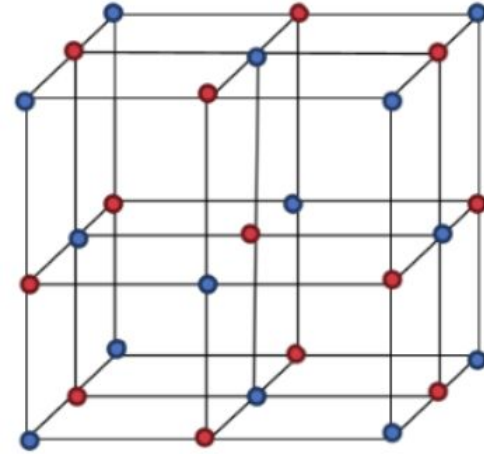
Quick Repetition:

- Crystal with ...



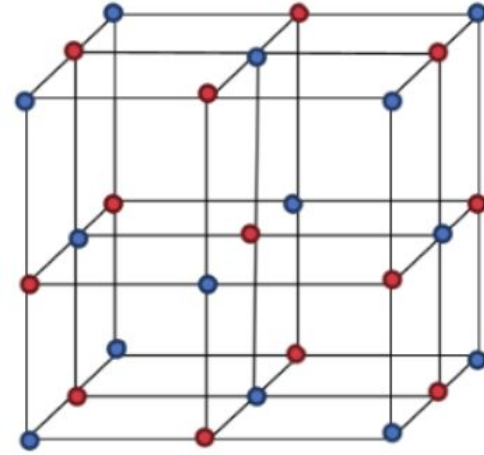
Quick Repetition:

- Crystal with ...
volume V



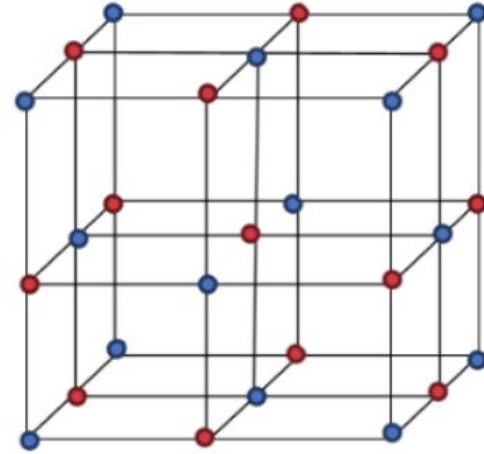
Quick Repetition:

- Crystal with ...
volume V
number of cells N_p



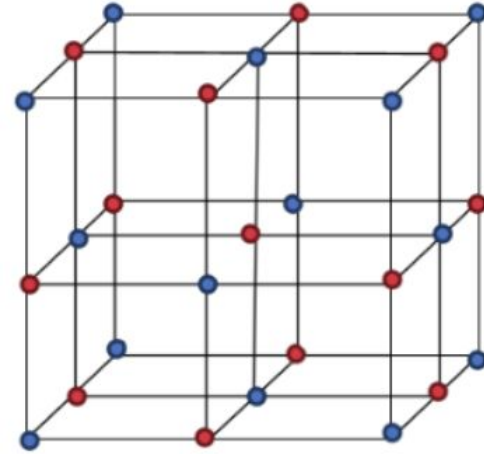
Quick Repetition:

- Crystal with ...
volume V
number of cells N_p
 r atoms in the unit-cell



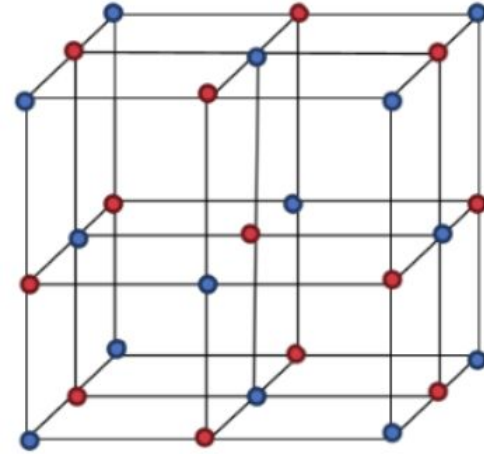
Quick Repetition:

- Crystal with ...
 - volume V
 - number of cells N_p
 - r atoms in the unit-cell
 - temperature T



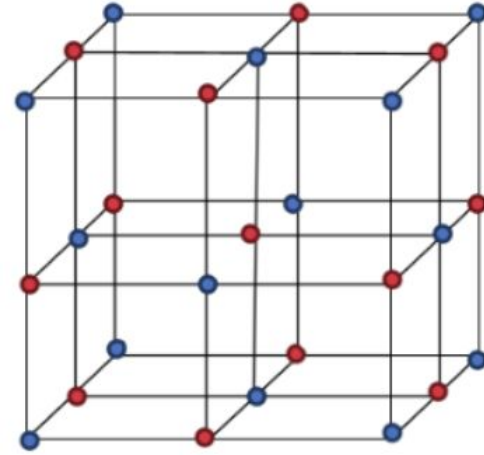
Quick Repetition:

- Crystal with ...
 - volume V
 - number of cells N_p
 - r atoms in the unit-cell
 - temperature T
- Born-Oppenheimer-Approximation (...)



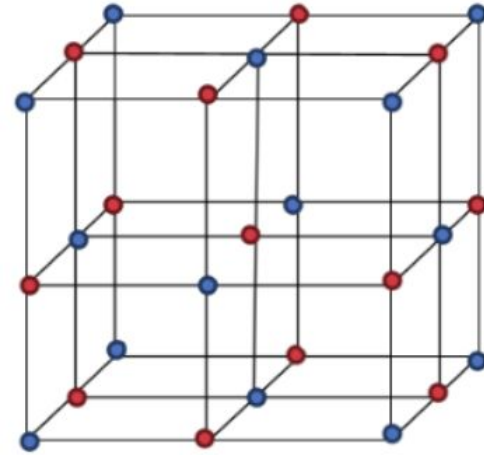
Quick Repetition:

- Crystal with ...
volume V
number of cells N_p
 r atoms in the unit-cell
temperature T
- Born-Oppenheimer-Approximation (...)
- Mean Field Approximation (...)



Quick Repetition:

- Crystal with ...
volume V
number of cells N_p
 r atoms in the unit-cell
temperature T
- Born-Oppenheimer-Approximation (...)
- Mean Field Approximation (...)



("reliable" for "many" observables & materials)

Notation 20.19 Within usual approximations, we describe the union of two non-interacting gases

$$H = H_{\text{el}} + H_{\text{ph}} = \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^{\dagger} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \Omega_{\mathbf{q},v} b_{\mathbf{q}v}^{\dagger} b_{\mathbf{q}v} \quad (20.26)$$

Notation 20.19 Within usual approximations, we describe the union of two non-interacting gases

$$H = H_{\text{el}} + H_{\text{ph}} = \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^{\dagger} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, \dots, r d_{\mathbf{x}}\}}} \Omega_{\mathbf{q}, \nu} b_{\mathbf{q}\nu}^{\dagger} b_{\mathbf{q}\nu} \quad (20.26)$$

$$\langle A \rangle \equiv \frac{\text{Tr}_{\mathcal{H}} A e^{-\beta(H_{\text{el}} - \mu N_{\text{el}} + H_{\text{ph}})}}{\text{Tr}_{\mathcal{H}} e^{-\beta(H_{\text{el}} - \mu N_{\text{el}} + H_{\text{ph}})}} \equiv \frac{\text{Tr}_{\mathcal{H}} A e^{-\beta(H_{\text{el}} - \mu N_{\text{el}} + H_{\text{ph}})}}{Z}.$$

$$\mathcal{H} = \mathcal{H}_{\text{F}} \otimes \mathcal{H}_{\text{B}}$$

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Fact 20.20 The expectation-value of the total internal energy can be computed as

$$\begin{aligned} \langle H \rangle_0 &= \langle H_{\text{el}} \rangle_0 + \langle H_{\text{ph}} \rangle_0 \\ &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} F_{\beta}(\epsilon_{m\mathbf{k}\sigma} - \mu) + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, r d_x\}}} \Omega_v(\mathbf{q}) B_{\beta}(\Omega_v(\mathbf{q})) \end{aligned} \quad (20.27)$$

Notation 21.1 We consider the interacting system that is described by the Hamiltonian from Definition 8.28 as

$$H = \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, \dots, rd_{\mathbf{x}}\}}} \hbar \omega_{\mathbf{q}, \nu} b_{\mathbf{q}\nu}^\dagger b_{\mathbf{q}\nu} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ \nu \in \{1, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger \right) \quad (21.1)$$

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Theorem 18.3 The electronic self-energy of this system reads can be approximated

$$\Sigma_{\substack{\mathbf{k}, \mathbf{k}' \\ m, m' \\ \mu, \mu'}}^{\text{Fock}}(i\omega_n) = \delta_{\mathbf{k}, \mathbf{k}'} \delta_{\mu, \mu'} \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \frac{g_{nmv}^*(\mathbf{k}, \mathbf{q}) g_{nm'v'}(\mathbf{k}, \mathbf{q})}{N_p} \left[\frac{B_\beta(\Omega_v(\mathbf{q})) + F_\beta(\xi_{\mathbf{q}+\mathbf{k}n\mu})}{i\hbar\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\mu} + \Omega_v(\mathbf{q})} + \frac{B_\beta(\Omega_v(\mathbf{q})) + 1 - F_\beta(\xi_{\mathbf{q}+\mathbf{k}n\mu})}{i\hbar\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\mu} - \Omega_v(\mathbf{q})} \right]$$

Notation 21.1 We consider the interacting system that is described by the Hamiltonian from Definition 8.28 as

$$H = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q},v}^\dagger b_{\mathbf{q},v} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q},m\sigma}^\dagger c_{\mathbf{k},n\sigma} \left(b_{\mathbf{q},v} + b_{-\mathbf{q},v}^\dagger \right) \quad (21.1)$$

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$$\mathcal{G}_{\substack{m, m' \\ \sigma, \sigma'}}(i\omega_n, \mathbf{k}) = \frac{\delta_{\sigma, \sigma'}}{\hbar} \left(\frac{1}{i\omega_n - (H_{0,\sigma}(\mathbf{k}) + \Sigma_\sigma(\mathbf{k}, i\omega_n) - \mu) / \hbar} \right)_{m, m'}$$

Notation 21.1 We consider the interacting system that is described by the Hamiltonian from Definition 8.28 as

$$\begin{aligned}
 H = & \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, \dots, rd_{\mathbf{x}}\}}} \hbar \omega_{\mathbf{q}, \nu} b_{\mathbf{q}\nu}^\dagger b_{\mathbf{q}\nu} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ \nu \in \{1, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger \right) \quad (21.1)
 \end{aligned}$$

Notation 21.1 We consider the interacting system that is described by the Hamiltonian from Definition 8.28 as

$$H = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q}v}^\dagger b_{\mathbf{q}v} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}v} + b_{-\mathbf{q}v}^\dagger \right) \quad (21.1)$$

$$\langle N \rangle = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \langle n_{m\mathbf{k}\sigma} \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E) \operatorname{Im} G_{m,m}^{\operatorname{ret}}(E + i\delta)_{\substack{\sigma, \sigma \\ \mathbf{k}, \mathbf{k}}}$$

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$$\langle H_{\text{cl}} \rangle = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \langle n_{m\mathbf{k}\sigma} \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_\beta(E) \operatorname{Im} G_{m,m}^{\operatorname{ret}}(E + i\delta, \mathbf{k})_{\substack{\sigma, \sigma}}$$

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$$\langle H_{\text{el}} \rangle = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \langle n_{m\mathbf{k}\sigma} \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_\beta(E) \operatorname{Im} G_{m,m}^{\operatorname{ret}}(E + i\delta, \mathbf{k})_{\substack{\sigma, \sigma}}$$

Problem (e.g. for the Phononic Contribution):

- the self-energy is no longer diagonal in $\mathbf{k}$
- the displacement Green-function D (with polarization Π) can in the interacting case no longer be related to the particle number

Fact 21.4 Going over to the field-integral-description of the problem gives the action

$$S[\chi, \overline{\chi}] = S_{\text{el}}[\psi, \overline{\psi}] + S_{\text{ph}}[\phi, \overline{\phi}] + S_{\text{el-ph-int}}[\chi, \overline{\chi}]$$

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with

$$\bullet \hspace{0.5cm} S_{\text{el}}[\psi,\overline{\psi}]=\sum_{\substack{\mathbf{k}\in\mathcal{Q} \\ \sigma\in\{\downarrow,\uparrow\} \\ \tau\in(0,\hbar\beta) \\ m\in\{1,2,\dots,\infty\}}} \overline{\psi}_{m\sigma}(\tau,\mathbf{k})\left(\partial_{\tau}+\frac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right)\psi_{m\sigma}(\tau,\mathbf{k})$$

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with

$$\begin{aligned} \bullet \quad S_{\text{el}}[\psi,\overline{\psi}]&= \sum_{\substack{\mathbf{k}\in Q \\ \sigma\in\{\downarrow,\uparrow\} \\ \tau\in(0,\hbar\beta) \\ m\in\{1,2,\dots,\infty\}}} \overline{\psi}_{m\sigma}(\tau,\mathbf{k}) \left(\partial_{\tau}+\frac{\epsilon_{m\sigma}(\mathbf{k})-\mu}{\hbar}\right) \psi_{m\sigma}(\tau,\mathbf{k}) \\ \\ \bullet \quad S_{\text{ph}}[\phi,\overline{\phi}]&= \sum_{\substack{\mathbf{q}\in Q \\ \tau\in(0,\hbar\beta) \\ \nu\in\{1,2,\dots,rd_{\mathbf{x}}\}}} \overline{\phi}_{\nu}(\tau,\mathbf{q}) \left(\partial_{\tau}+\omega_{\nu}(\mathbf{q})\right) \phi_{\nu}(\tau,\mathbf{q}) \end{aligned}$$

Fact 21.4 Going over to the field-integral-description of the problem gives the action

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Fact 21.4 Going over to the field-integral-description of the problem gives the action

$$S[\chi, \bar{\chi}] = S_{\text{cl}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{cl-ph-int}}[\chi, \bar{\chi}]$$

with

- $S_{\text{cl}}[\psi, \bar{\psi}] = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(\tau, \mathbf{k}) \left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(\tau, \mathbf{k})$
- $S_{\text{ph}}[\phi, \bar{\phi}] = \sum_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(\tau, \mathbf{q}) (\partial_\tau + \omega_v(\mathbf{q})) \phi_v(\tau, \mathbf{q})$
- $S_{\text{cl-ph-int}}[\chi, \bar{\chi}] = \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, 2, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \left(\phi_v(\tau, \mathbf{q}) + \bar{\phi}_v(\tau, -\mathbf{q}) \right) \bar{\psi}_{m\sigma}(\tau, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\tau, \mathbf{k})$

Notation 21.5 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{cl}}, \bar{j}_{\text{cl}}, j_{\text{ph}}, \bar{j}_{\text{ph}}] \equiv \int \mathcal{D}(\bar{\chi}, \chi) \exp \left[- \left(S_{\text{cl}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{cl-ph-int}}[\chi, \bar{\chi}] \right) + \sum_{\bullet \in \{\text{cl}, \text{ph}\}} \bar{j}_{\bullet, \chi_{\bullet}} + \bar{\chi}_{\bullet, j_{\bullet}} \right] \quad (21.6)$$

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$$S[\chi, \bar{\chi}] = S_{\text{cl}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{cl-ph-int}}[\chi, \bar{\chi}]$$

with

- $S_{\text{cl}}[\psi, \bar{\psi}] = \oint \limits_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(\tau, \mathbf{k}) \left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(\tau, \mathbf{k})$
- $S_{\text{ph}}[\phi, \bar{\phi}] = \oint \limits_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_\nu(\tau, \mathbf{q}) (\partial_\tau + \omega_\nu(\mathbf{q})) \phi_\nu(\tau, \mathbf{q})$
- $S_{\text{cl-ph-int}}[\chi, \bar{\chi}] = \oint \limits_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, 2, \dots, \infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \left(\phi_\nu(\tau, \mathbf{q}) + \bar{\phi}_\nu(\tau, -\mathbf{q}) \right) \bar{\psi}_{m\sigma}(\tau, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\tau, \mathbf{k})$

Notation 21.5 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{cl}}, \bar{j}_{\text{cl}}, j_{\text{ph}}, \bar{j}_{\text{ph}}] \equiv \int \mathcal{D}(\bar{\chi}, \chi) \exp \left[- \left(S_{\text{cl}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{cl-ph-int}}[\chi, \bar{\chi}] \right) + \sum_{\bullet \in \{\text{cl}, \text{ph}\}} \bar{j}_{\bullet, \chi} + \bar{\chi}_{\bullet, j} \right] \quad (21.6)$$

with abbreviations

- $\bar{j}_{\text{cl}} \chi_{\text{cl}} \equiv \oint \limits_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{j}_{m\sigma}(\tau, \mathbf{k}) \psi_{m\sigma}(\tau, \mathbf{k})$
- $\bar{j}_{\text{ph}} \chi_{\text{ph}} \equiv \oint \limits_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{j}_\nu(\tau, \mathbf{q}) \phi_\nu(\tau, \mathbf{q})$
- etc...

Lemma 21.16 For the phononic occupation numbers of the interacting system we have the identity

$$\langle H_{\text{ph}} \rangle \approx -\frac{1}{\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \mathcal{G}_{0,v}(q)$$

Corollary 21.17 Evaluating the Matsubara-sums simplifies the individual terms as

$$\langle H_{\text{ph}} \rangle \approx \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \Omega_v(\mathbf{q}) B_\beta(\Omega_v(\mathbf{q}))$$

Lemma 21.16 For the phononic occupation numbers of the interacting system we have the identity

$$\begin{aligned}
\langle H_{\text{ph}} \rangle &\approx -\frac{1}{\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \mathcal{G}_{0,v}(q) \\
&+ \frac{1}{\beta^2 N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n' \in \{1, \dots, \infty\} \\ v \in \{1,2,\dots,rd_x\}}} \Omega_v(\mathbf{q} = \mathbf{0}) g_{nnv}(\mathbf{k}, 0) g_{n'n'v}(\mathbf{k}', 0) \mathcal{G}_{0,v}^2(q = 0) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma'}(k') \\
&- \frac{1}{\beta^2 N_p} \sum_{\substack{k \in i\Omega_{-1} \times Q \\ q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \Omega_v(\mathbf{q}) |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 \mathcal{G}_{0,v}^2(q) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma}(k+q) + \mathcal{O}(g^4)
\end{aligned}$$

Corollary 21.17 Evaluating the Matsubara-sums simplifies the individual terms as

$$\begin{aligned}
\langle H_{\text{ph}} \rangle &\approx \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \Omega_v(\mathbf{q}) B_\beta(\Omega_v(\mathbf{q})) + \sum_{v \in \{1,2,\dots,rd_x\}} A_v B_v^2 \\
&- \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1,2,\dots,rd_x\} \\ n, n' \in \{1, \dots, \infty\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 (F_\beta(\xi_{n,\sigma,\mathbf{k}}) - F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})) \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}})^2} + \frac{1}{4} \frac{\beta \text{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}} \right] \\
&+ \mathcal{O}(g^4)
\end{aligned}$$

with real coefficients A_v and B_v defined as

$$A_v = \Omega_v(\mathbf{q} = \mathbf{0}) \mathcal{G}_{0,v}^2(q = 0) = \Omega_v^{-1}(\mathbf{q} = \mathbf{0}), \quad B_v = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, \dots, \infty\}}} g_{nnv}(\mathbf{k}, 0) F_\beta(\xi_{n\mathbf{k}\sigma}) \quad (21.24)$$

$$\begin{aligned}
H &= H_0 + V \\
&= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q}v}^\dagger b_{\mathbf{q}v} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}v} + b_{-\mathbf{q}v}^\dagger \right)
\end{aligned}$$

Lemma 8.28 The hermiticity of the Hamiltonian implies for the coupling

$$g_{mnv}(\mathbf{k}, \mathbf{q}) = g_{nmv}^*(\mathbf{k} + \mathbf{q}, -\mathbf{q}).$$

Lemma 8.29 Notice that for inversion-symmetric crystals one has the additional symmetry

$$g_{mnv}(\mathbf{k}, \mathbf{q}) = g_{mnv}^*(-\mathbf{k}, -\mathbf{q}).$$

Workflow of the EPW-Project

CIF-File

Python Step 0 Automat

Input-files for scf, relaxation

Python Dardel

Relaxed Structural Parameters

Python Step 1 Automat

Input-files for scf, bands, DOS, PDOS

Python Dardel

Output-files → Evaluation (Python)

Python Step 2 Automat

Input-files for scf, nsf, Wannier90

Python Dardel

Output-files → Comparison w. QE-BSE

Python Step 3 Automat

Input-files for scf, DFPT, Fourier, EPW-Bands, EPW-Couplings

Python Dardel

Output-files → Evaluation: Comparison of Bands & Band Symmetries

$$H = H_0 + V$$

$$= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q}v}^\dagger b_{\mathbf{q}v} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} (b_{\mathbf{q}v} + b_{-\mathbf{q}v}^\dagger)$$

Lemma 8.28 The hermiticity of the Hamiltonian implies for the coupling

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Workflow of the EPW-Project

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Output-files → Evaluation (Python)

Python Step 2 Automat

Input-files for scf, nsf, Wannier90

Python Dardel

Output-files → Comparison w. QE-BS

Python Step 3 Automat

Input-files for scf, DFPT, Fourier, EPW-Bands, EPW-Couplings

Python Dardel

Output-files → Evaluation: Comparison of Bands & Band Symmetries

Input-files Fortran (GRIT) → Specific Heat vs T.
Dardel

```
|| GGGGGGGGGGGG RRRRRRRRRR IIIIIIIIIII TTTTTTTTTTTTTT ||
|| GGGGGGGGGGGG RRRRRRRRRR IIIIIIIIIII TTTTTTTTTTTTTT ||
|| GGG          RRRR    RRRR      IIII      TTTT      ||
|| GGG          RRRR    RRRR      IIII      TTTT      ||
|| GGG    GGGG  RRRRRRRRRR      IIII      TTTT      ||
|| GGG    GGGG  RRRRRRRRRR      IIII      TTTT      ||
|| GGG    GG   RRRR    RRRR      IIII      TTTT      ||
|| GGG    GG   RRRR    RRRR      IIII      TTTT      ||
|| GGGGGGGGGGGG RRRR    RRRR  IIIIIIIIIII TTTT      ||
|| GGGGGGGGGGGG RRRR      RRR  IIIIIIIIIII TTTT      ||
||
```

[Gr]een-Functions for [I]nteracting [T]hermodynamics
(Post-Processing Tool for the Electron-Phonon-Calculations)

```
+=====+
|| Information about Input-Parameters ||
+=====+
```

```
+-----+
| Modes |
+-----+
```

```
- Fine k-grid near Fermi surface = F
- Fine q-grid near the Gamma-point = F
- Interacting Electrons and phonons = T
- Debug mode enabled = F
```

```
+-----+
| Tasks |
+-----+
```

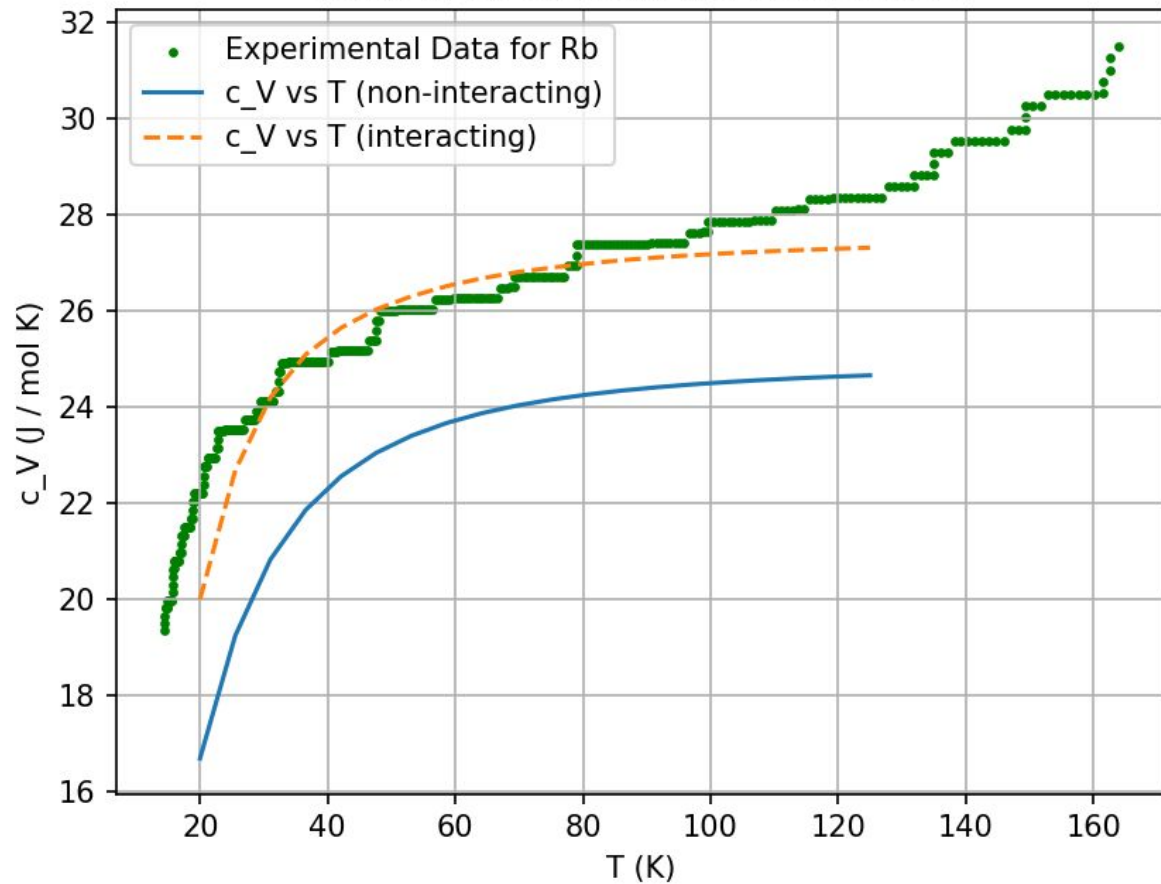
```
- Create binary files = T
- Calc N_states_per_UC (mu = E_F, T = 0) = T
- Calc N_per_UC (mu = E_F, T = 0) = T
- Calc mu vs Temperature = T
- Calc specific heat per UC = T
- Calc Sommerfeld gamma from linear fit = F
```

```
+-----+
| Temperature parameters |
+-----+
```

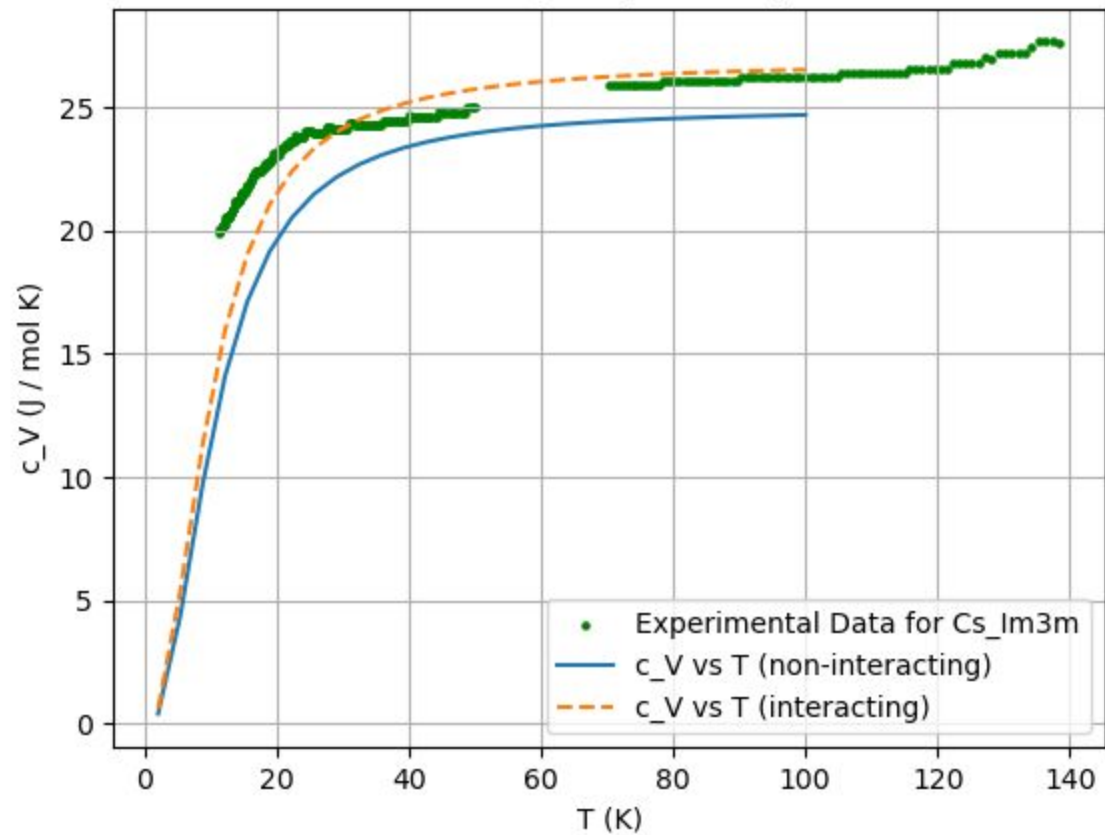
```
- Number of temperature points = 20
- Minimum temperature (K) = 20.0000
- Maximum temperature (K) = 125.0000
```

Results:

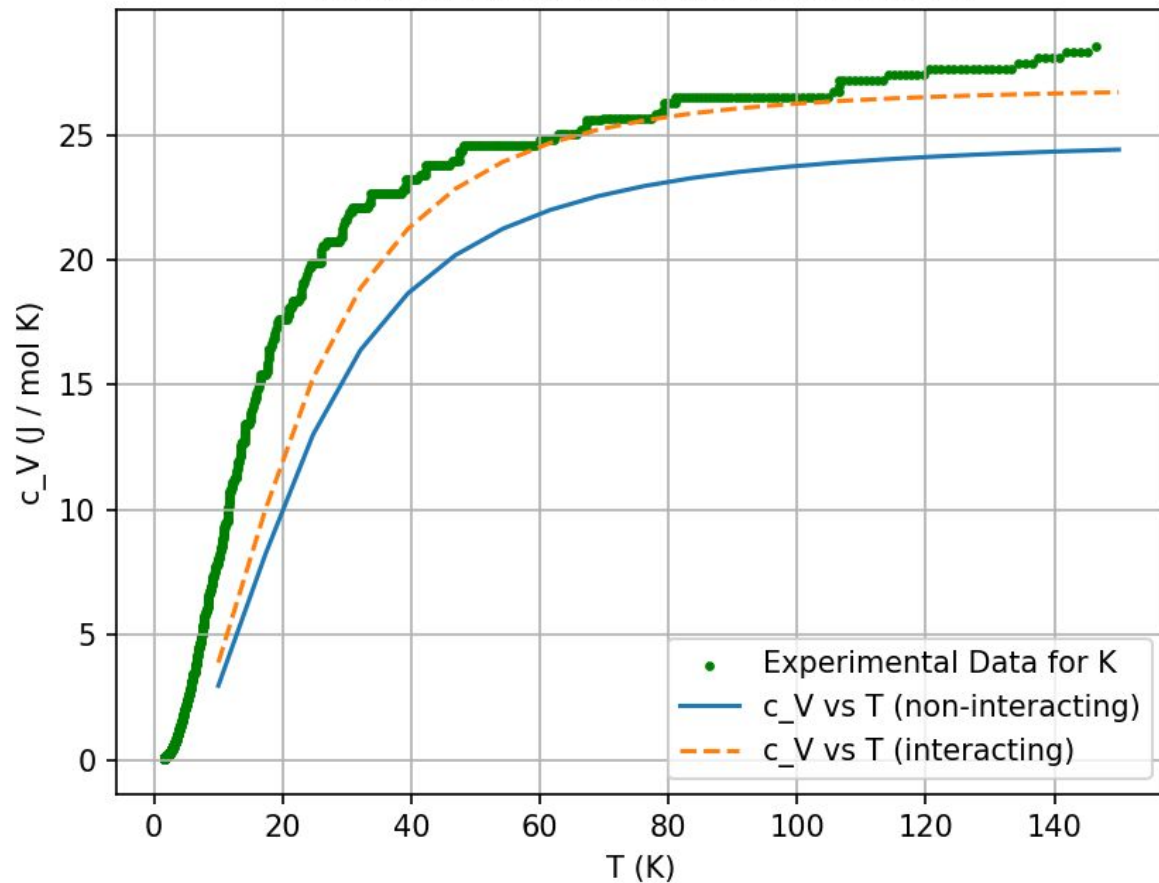
Molar Heat Capacity vs. Temperature



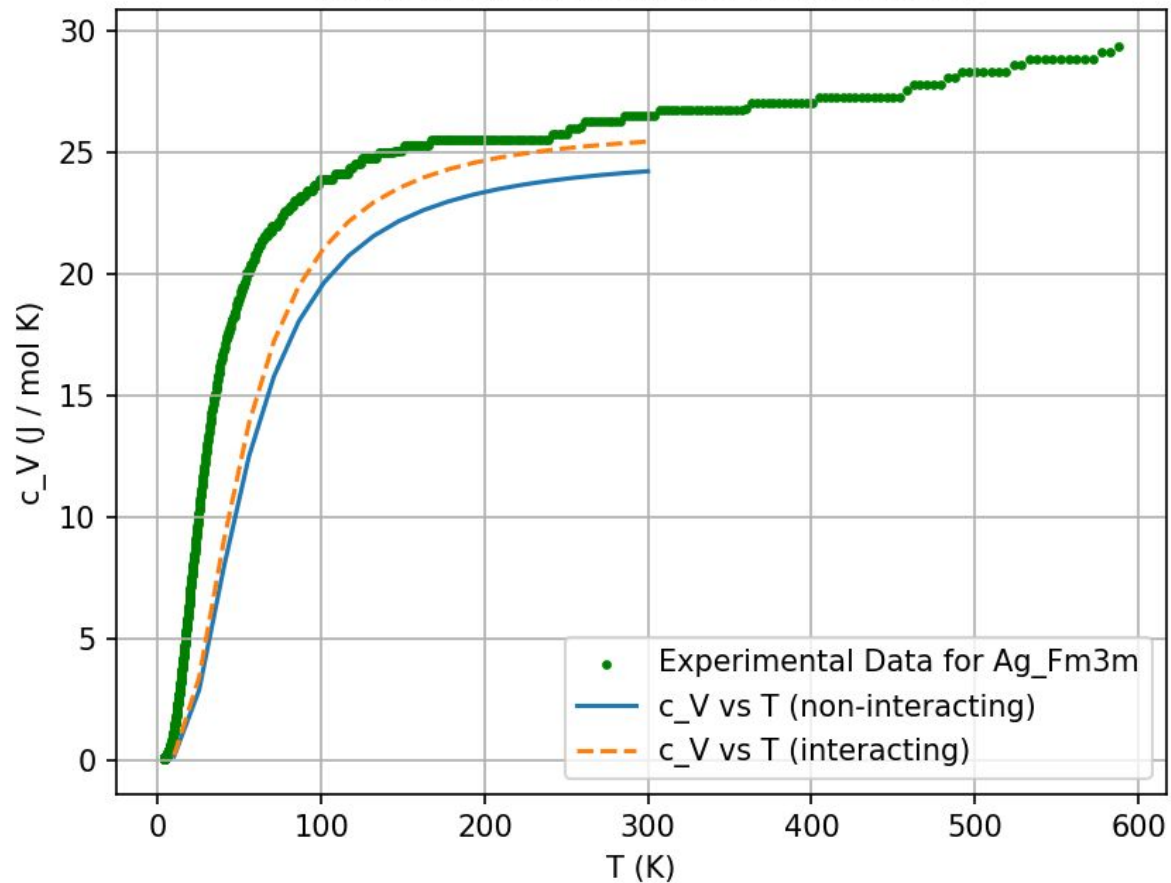
Molar Heat Capacity vs. Temperature



Molar Heat Capacity vs. Temperature



Molar Heat Capacity vs. Temperature



Interpretation:

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$$\begin{aligned}
 \langle H_{\text{ph}} \rangle \approx & \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \Omega_{\nu}(\mathbf{q}) B_{\beta}(\Omega_{\nu}(\mathbf{q})) + \sum_{\nu \in \{1,2,\dots,rd_{\mathbf{x}}\}} A_{\nu} B_{\nu}^2 \\
 & - \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\} \\ n, n' \in \{1,\dots,\infty\}}} \frac{\Omega_{\nu}(\mathbf{q})}{N_p} |g_{n'n\nu}(\mathbf{k}, \mathbf{q})|^2 (F_{\beta}(\xi_{n,\sigma,\mathbf{k}}) - F_{\beta}(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})) \left[\frac{B_{\beta}(\Omega_{\nu}(\mathbf{q})) - B_{\beta}(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\Omega_{\nu}(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_{\nu}(\mathbf{q})/2)^2}{\Omega_{\nu}(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}} \right] \\
 & + \mathcal{O}(g^4) \\
 \equiv & \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \Omega_{\nu}(\mathbf{q}) B_{\beta}(\Omega_{\nu}(\mathbf{q})) + \sum_{\nu \in \{1,2,\dots,rd_{\mathbf{x}}\}} A_{\nu} B_{\nu}^2 + \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1,\dots,\infty\}}} K_{\mathbf{k},\sigma,n} + \mathcal{O}(g^4)
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 & - \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ n, n' \in \{1, \dots, \infty\}}} \frac{\Omega_{\nu}(\mathbf{q})}{N_p} |g_{n'n\nu}(\mathbf{k}, \mathbf{q})|^2 (F_{\beta}(\xi_{n, \sigma, \mathbf{k}}) - F_{\beta}(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}})) \left[\frac{B_{\beta}(\Omega_{\nu}(\mathbf{q})) - B_{\beta}(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_{\nu}(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_{\nu}(\mathbf{q})/2)^2}{\Omega_{\nu}(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}}} \right] \\
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 \end{aligned}$$

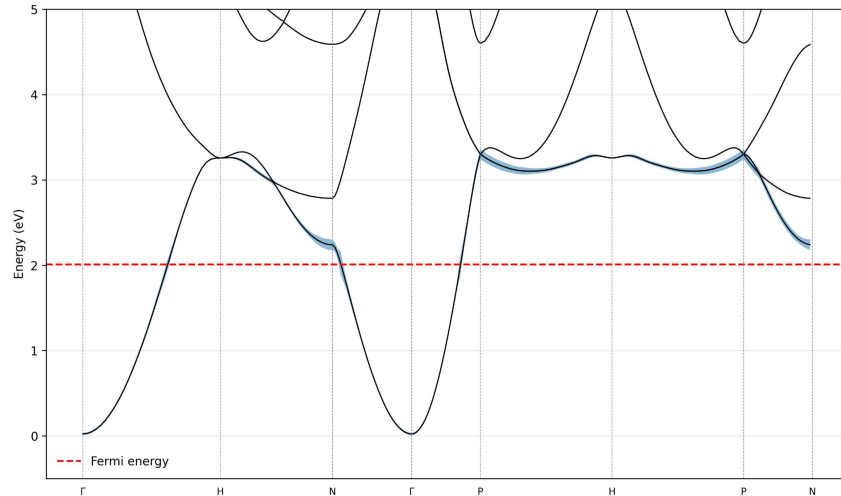
Interpretation:

$$\begin{aligned}
 \langle H_{\text{ph}} \rangle \approx & \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \Omega_{\nu}(\mathbf{q}) B_{\beta}(\Omega_{\nu}(\mathbf{q})) + \sum_{\nu \in \{1, 2, \dots, rd_{\mathbf{x}}\}} A_{\nu} B_{\nu}^2 \\
 & - \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ n, n' \in \{1, \dots, \infty\}}} \frac{\Omega_{\nu}(\mathbf{q})}{N_p} |g_{n'n\nu}(\mathbf{k}, \mathbf{q})|^2 (F_{\beta}(\xi_{n, \sigma, \mathbf{k}}) - F_{\beta}(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}})) \left[\frac{B_{\beta}(\Omega_{\nu}(\mathbf{q})) - B_{\beta}(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_{\nu}(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_{\nu}(\mathbf{q})/2)^2}{\Omega_{\nu}(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}}} \right] \\
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 \end{aligned}$$

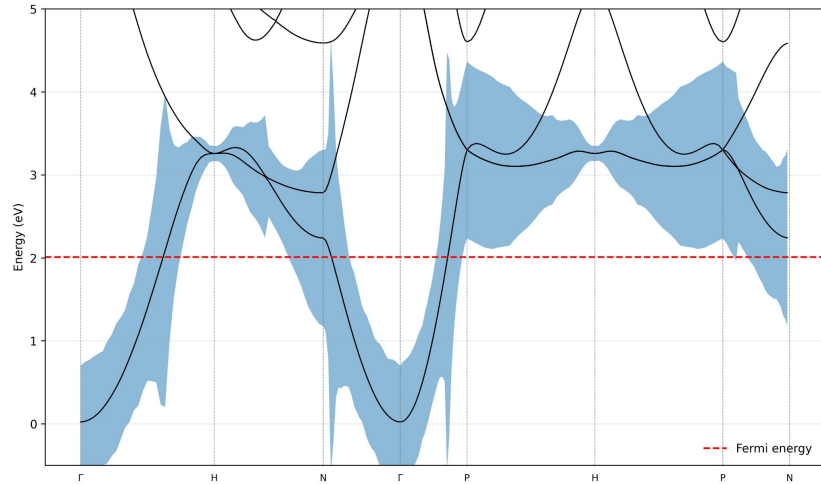
Let's take a closer look at the kernel!

Interpretation Electronically Resolved Kernel at

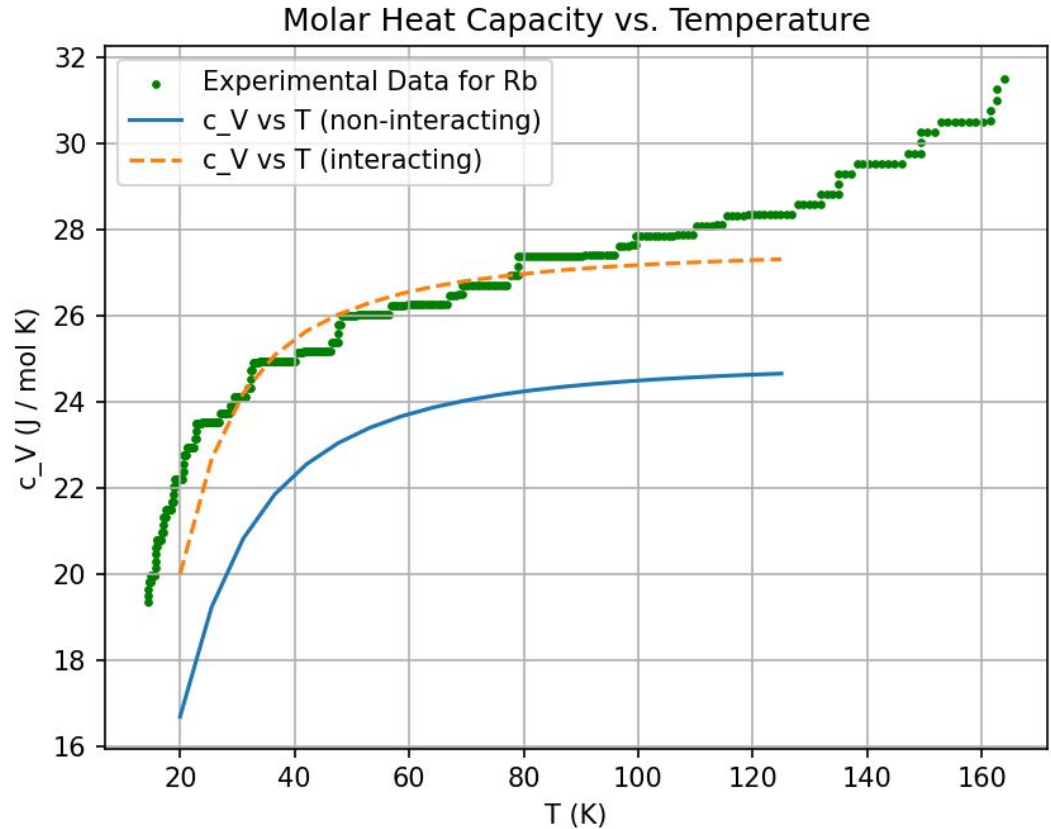
$T = 10\text{ K}$



$T = 100\text{ K}$



The reason why it doesn't increase with increasing temperature:



Reason:

Why computed specific heat at constant volume, but in experiments, one measures at constant pressure!

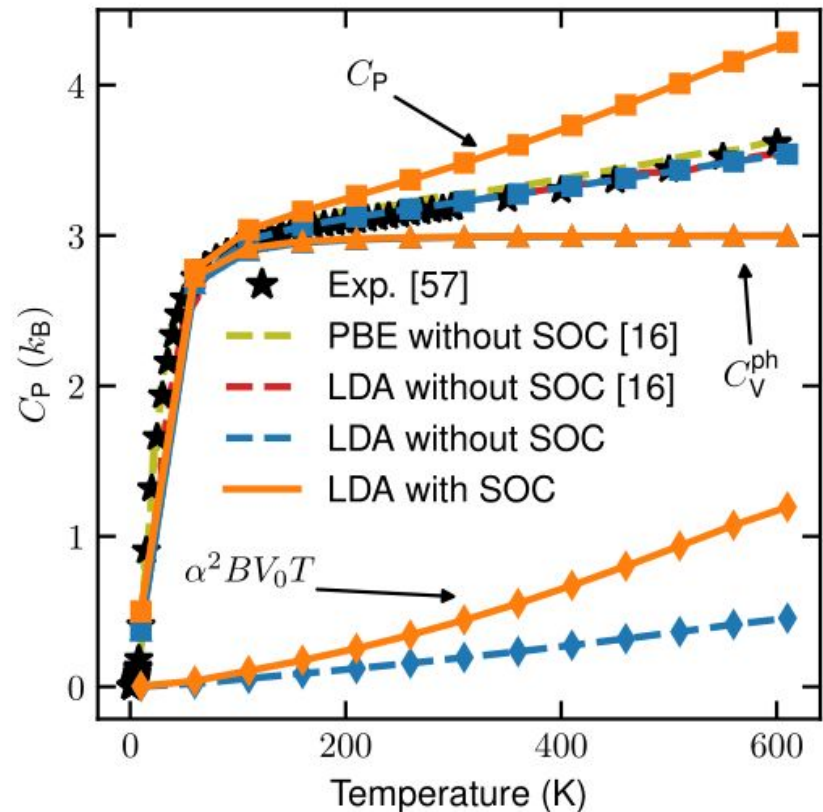
Effects of Spin-Orbit Coupling and Thermal Expansion on the Phonon-limited Resistivity of Pb from First Principles

Félix Antoine Goudreault,¹ Samuel Poncé,^{2,3} Feliciano Giustino,^{4,5} and Michel Côté¹

volume $V_0(T)$. Most experiments are instead made at constant pressure which can be obtained as [30]:

$$C_P(T) = C_V(T) + \alpha^2 B V_0(T) T, \quad (5)$$

where B is taken from the free energy fits of Eq. (3) and $\alpha(T) = \frac{1}{V_0} \frac{\partial V_0(T)}{\partial T} \big|_P$ is the linear TE coefficient, evaluated at constant pressure P .



4 / 5 : Fermi-Surfaces

[Currently still a “draft”]

Tensor-Moment Analysis of Spin-Orbit Coupling Enhancement through Electronic Interactions

Sebastian Kalhöfer  * and Lars Nordström 

Department of Physics and Astronomy, Uppsala University, Box 516, 75120 Uppsala, Sweden

Bloch-Kohn-Sham-**Wavefunctions** for a single electron in a crystal:

$$(H\Psi_{sv})(\mathbf{r}, s) = E_{sv}\Psi_{sv}(\mathbf{r}, s),$$

Bloch-Kohn-Sham-**Wavefunctions** for a single electron in a crystal:

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- are being used to extract ground-state-properties through observables:

Bloch-Kohn-Sham-**Wavefunctions** for a single electron in a crystal:

$$(H\Psi_{sv})(\mathbf{r}, s) = E_{sv}\Psi_{sv}(\mathbf{r}, s),$$

- are being used to extract ground-state-properties through observables:

Definition 9.1 The *[F]ermi-[S]urface* of a non-interacting metallic system is defined as the set

$$\text{FS} = \left\{ \mathbf{k} \in \mathbf{Q} \mid \exists n \in \mathbb{N} : \epsilon_{n\mathbf{k}} = \mu(T = 0 \text{ K}) \equiv E_{\text{F}} \right\}.$$

Example-System:

- Sr_2RhO_4 (Strontium rhodate)

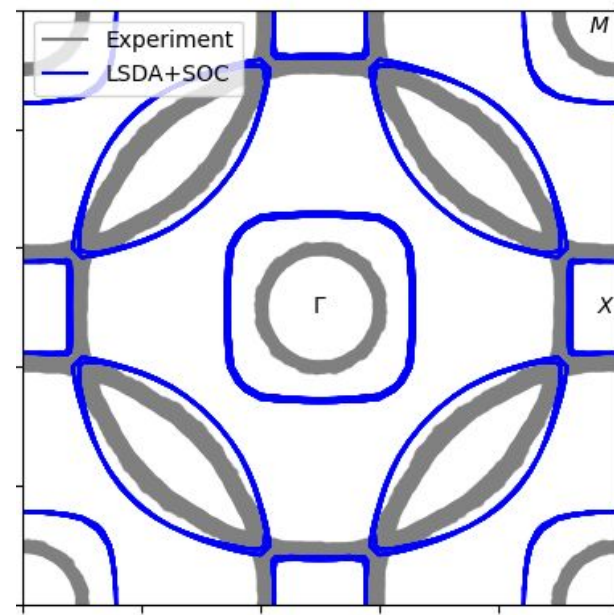
```
/home/sebastian/Dropbox/AA_Sebast-d_distorted_Structure_3p8K/elk.in [----] 0 L:[ 1+ 0 1/ 45]
avec
  1.0000000000 0.0000000000 0.0000000000
  0.0000000000 1.0000000000 0.0000000000
  0.5000000000 0.5000000000 2.3709245679

scale
10.2718329837

atoms
3
: nspecies
'Rh.in' : spfname
4
0.000000000000000 0.000000000000000 0.000000000000000 0.00000000 0.00000000 0.00000000
0.750000000000000 0.250000000000000 0.500000000000000 0.00000000 0.00000000 0.00000000
0.250000000000000 0.750000000000000 0.500000000000000 0.00000000 0.00000000 0.00000000
0.500000000000000 0.500000000000000 0.000000000000000 0.00000000 0.00000000 0.00000000
'Sr.in' : spfname
8
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0.925910000000000 0.425910000000000 0.148180000000000 0.00000000 0.00000000 0.00000000
0.574090000000000 0.074090000000000 0.851820000000000 0.00000000 0.00000000 0.00000000
0.425910000000000 0.925910000000000 0.148180000000000 0.00000000 0.00000000 0.00000000
0.074090000000000 0.574090000000000 0.851820000000000 0.00000000 0.00000000 0.00000000
0.175910000000000 0.175910000000000 0.648180000000000 0.00000000 0.00000000 0.00000000
0.675910000000000 0.675910000000000 0.648180000000000 0.00000000 0.00000000 0.00000000
0.324090000000000 0.324090000000000 0.351820000000000 0.00000000 0.00000000 0.00000000
'O.in' : spfname
16
0.952500000000000 0.952500000000000 0.500000000000000 0.00000000 0.00000000 0.00000000
0.797500000000000 0.297500000000000 0.000000000000000 0.00000000 0.00000000 0.00000000
0.702500000000000 0.797500000000000 0.000000000000000 0.00000000 0.00000000 0.00000000
0.297500000000000 0.202500000000000 0.000000000000000 0.00000000 0.00000000 0.00000000
0.047500000000000 0.452500000000000 0.500000000000000 0.00000000 0.00000000 0.00000000
0.547500000000000 0.547500000000000 0.500000000000000 0.00000000 0.00000000 0.00000000
0.202500000000000 0.702500000000000 0.000000000000000 0.00000000 0.00000000 0.00000000
0.452500000000000 0.047500000000000 0.500000000000000 0.00000000 0.00000000 0.00000000
0.919700000000000 0.919700000000000 0.160600000000000 0.00000000 0.00000000 0.00000000
0.830300000000000 0.330300000000000 0.339400000000000 0.00000000 0.00000000 0.00000000
0.669700000000000 0.169700000000000 0.660600000000000 0.00000000 0.00000000 0.00000000
0.330300000000000 0.830300000000000 0.339400000000000 0.00000000 0.00000000 0.00000000
0.169700000000000 0.669700000000000 0.660600000000000 0.00000000 0.00000000 0.00000000
0.080300000000000 0.080300000000000 0.839400000000000 0.00000000 0.00000000 0.00000000
0.580300000000000 0.580300000000000 0.839400000000000 0.00000000 0.00000000 0.00000000
0.419700000000000 0.419700000000000 0.160600000000000 0.00000000 0.00000000 0.00000000
```

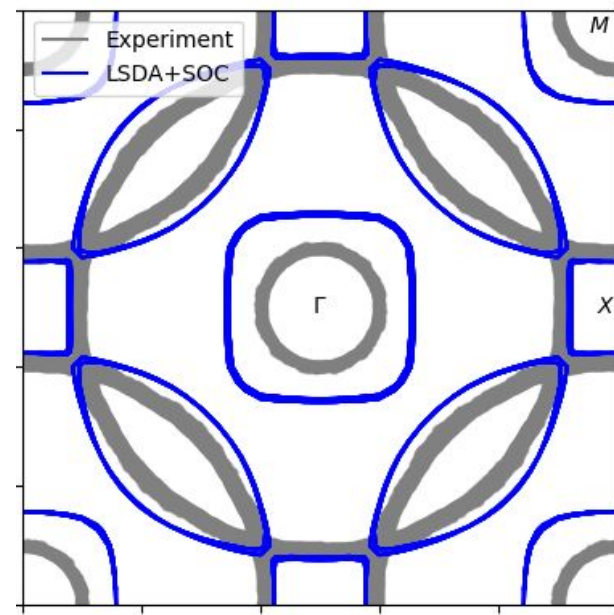
LDA (ELK-9.5.14 Calculation)

LDA (ELK-9.5.14 Calculation)



LDA (ELK-9.5.14 Calculation)

... “behind the scenes” (...)



Fact:

Sometimes the “independent electrons”-approximation fails to reproduce experimental results

→ DFT+U

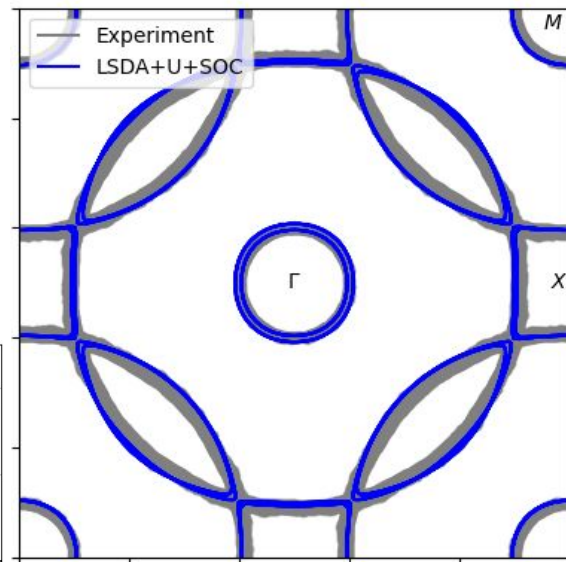
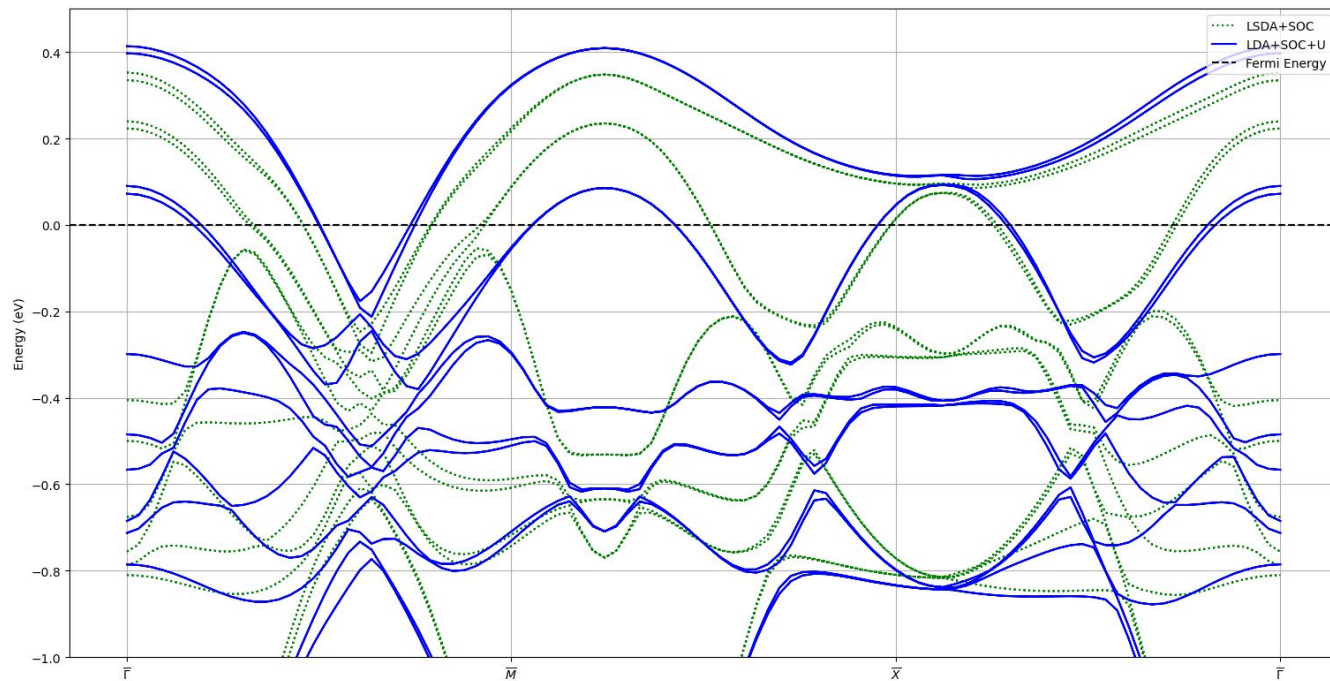
Planewaves, Pseudopotentials, and the LAPW Method

Second Edition

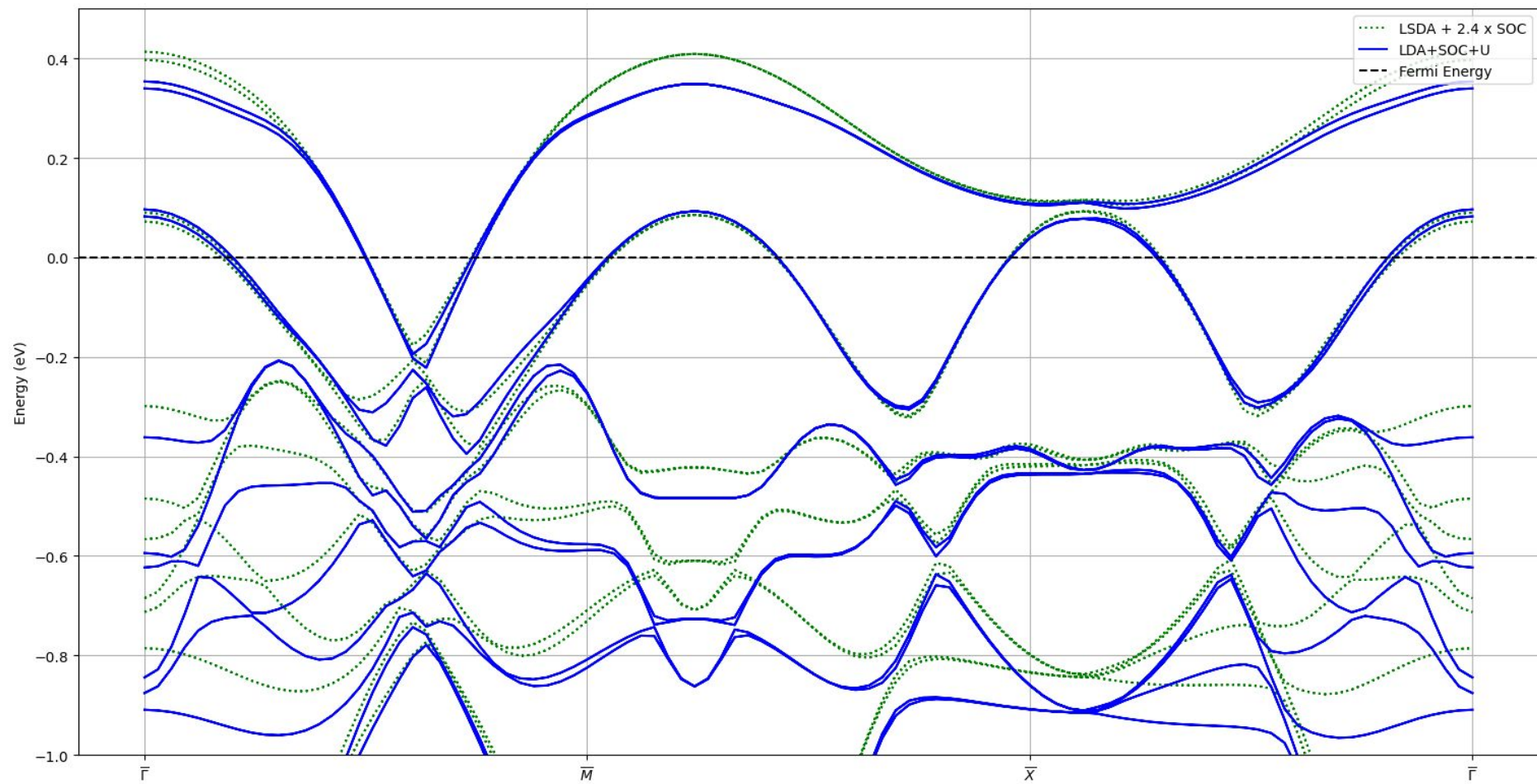
Edited by
David J. Singh
Lars Nordstrom

DFT+U (ELK-9.5.14 Calculation):

- Yukawa-type of screening with $U = 3.6$ eV
- Around Mean Field Approximation



Notice:



Recap:

Notation 6.1 We consider the following sequence of Hilbert-spaces and densities

$$\rho_{\mathcal{F}} = \frac{1}{Z} \mathrm{e}^{-\beta (H^{2Q} - \mu N^{2Q})} \qquad \in \mathcal{D} \Big[\quad \mathcal{H} = \mathcal{F}(\mathcal{H}) \quad \Big]$$

Theorem 6.2 For any one-body operator X on $\mathcal{H}$ one finds for the expectation-values

$$\begin{aligned} \text{"Observable Quantity"} &= \langle X^{2Q} \rangle \\ &= \mathrm{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right) \end{aligned}$$

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$$\begin{aligned} \rho_{\mathcal{F}} &= \frac{1}{Z} e^{-\beta(H^{2Q} - \mu N^{2Q})} && \in \mathcal{D} \left[\begin{array}{c} \mathcal{H} = \mathcal{F}(\mathcal{H}) \\ \cup \\ \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in Q} \mathcal{H}_{\mathbf{k}} \end{array} \right] \\ \rho_{\mathcal{H}} &= \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}}\rangle \langle \Psi_{N\mathbf{k}}^{\text{sv}}| && \in \mathcal{D} \left[\begin{array}{c} \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in Q} \mathcal{H}_{\mathbf{k}} \end{array} \right] \end{aligned}$$

Theorem 6.2 For any one-body operator X on $\mathcal{H}$ one finds for the expectation-values

"Observable Quantity" = $\langle X^{2Q} \rangle$
= $\text{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right)$
= $\text{Tr}_{\mathcal{H}} \left(\rho_{\mathcal{H}} X \right)$

= $\sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \left\langle \Psi_{N\mathbf{k}}^{\text{sv}} \left| X \right| \Psi_{N\mathbf{k}}^{\text{sv}} \right\rangle$

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 & \bigcup & \\
 \rho_{\mathcal{H}} = \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}} \rangle \langle \Psi_{N\mathbf{k}}^{\text{sv}}| & \in \mathcal{D} \Big[& \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in Q} \mathcal{H}_{\mathbf{k}} \quad \Big] \\
 & \bigcup & \\
 \rho_{N\mathbf{k}} = |u_{N\mathbf{k}}^{\text{sv}} \rangle \langle u_{N\mathbf{k}}^{\text{sv}}| & \in \mathcal{D} \Big[& \mathcal{H}_{\mathbf{k}} = L^2(\text{UC}) \otimes \mathbb{C}^2 \quad \Big]
 \end{array}$$

Theorem 6.2 For any one-body operator X on $\mathcal{H}$ one finds for the expectation-values

$$\begin{aligned}
 \text{"Observable Quantity"} &= \langle X^{2Q} \rangle \\
 &= \text{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right) \\
 &= \text{Tr}_{\mathcal{H}} \left(\rho_{\mathcal{H}} X \right) &= \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \left\langle \Psi_{N\mathbf{k}}^{\text{sv}} \right| X \left| \Psi_{N\mathbf{k}}^{\text{sv}} \right\rangle \\
 &= N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \text{Tr}_{\mathcal{H}_{\mathbf{k}}} \left(\rho_{N\mathbf{k}} X \right) &= N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \left\langle u_{N\mathbf{k}}^{\text{sv}} \right| X \left| u_{N\mathbf{k}}^{\text{sv}} \right\rangle_{\text{UC}}
 \end{aligned}$$

Fact:

For *some* observables, there is an *extended* decomposition, that allows us extract even more information!

$$\rho_{\mathcal{F}} = \frac{1}{Z} \mathrm{e}^{-\beta(H^{2Q} - \mu N^{2Q})} \qquad \in \mathcal{D} \left[\begin{array}{l} \mathcal{H} = \mathcal{F}(\mathcal{H}) \end{array} \right]$$

$$\text{"Observable Quantity"} = \langle X^{2Q} \rangle = \mathrm{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right) = \mathrm{Tr}_H \left(\rho_H X \right) = \dots$$

$$\langle X^{2Q} \rangle = \langle X^{2Q} \rangle_{\mathcal{F}} = \langle X^{2Q} \rangle_H$$

$$\rho_{\mathcal{F}} = \frac{1}{Z} \mathrm{e}^{-\beta (H^{2Q} - \mu N^{2Q})} \qquad \in \mathcal{D} \left[\begin{array}{c} \mathcal{H} = \mathcal{F}(H) \end{array} \right]$$

$$\bigcup$$

$$\rho_{\mathcal{H}} = \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}} \rangle \langle \Psi_{N\mathbf{k}}^{\text{sv}}| \qquad \in \mathcal{D} \left[\begin{array}{c} \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in Q} \mathcal{H}_{\mathbf{k}} \end{array} \right]$$

$$\text{"Observable Quantity"} = \langle X^{2Q} \rangle = \text{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right) = \text{Tr}_H \left(\rho_H X \right) = \dots$$

$$\dots = N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \text{Tr}_{\mathcal{H}_{\mathbf{k}}} \left(\rho_{N\mathbf{k}} X \right)$$

$$\begin{array}{ll} \rho_{\mathcal{F}} = \frac{1}{Z} \mathrm{e}^{-\beta(H^{2Q} - \mu N^{2Q})} & \in \mathcal{D} \left[\begin{array}{c} \mathcal{H} = \mathcal{F}(\mathcal{H}) \\ \bigcup \\ \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in \mathbb{Q}} \mathcal{H}_{\mathbf{k}} \\ \bigcup \\ \mathcal{H}_{\mathbf{k}} = L^2(\text{UC}) \otimes \mathbb{C}^2 = \left[\left(\bigoplus_{\alpha=1}^r L^2([0, R_a], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \right) \oplus \left(L^2(\mathbb{IR}) \otimes \mathbb{C}^2 \right) \right] \end{array} \right] \\ \rho_{\mathcal{H}} = \sum_{\mathbf{k} \in \mathbb{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}} \rangle \langle \Psi_{N\mathbf{k}}^{\text{sv}}| & \\ \rho_{N\mathbf{k}} = |u_{N\mathbf{k}}^{\text{sv}} \rangle \langle u_{N\mathbf{k}}^{\text{sv}}| & \end{array}$$

$$\text{"Observable Quantity"} = \langle X^{2Q} \rangle = \text{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right) = \text{Tr}_{\mathcal{H}} \left(\rho_{\mathcal{H}} X \right) = \dots$$

$$\begin{aligned} \dots &= N_p \sum_{\mathbf{k} \in \mathbb{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \text{Tr}_{\mathcal{H}_{\mathbf{k}}} \left(\rho_{N\mathbf{k}} X \right) \\ &= N_p \sum_{\mathbf{k} \in \mathbb{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \text{Tr}_{\mathcal{H}_a} \left(\rho_{N\mathbf{k}}^{\alpha} X \right) \end{aligned}$$

$$\begin{array}{ll} \rho_{\mathcal{F}} = \frac{1}{Z} \mathrm{e}^{-\beta(H^{2Q} - \mu N^{2Q})} & \in \mathcal{D} \left[\begin{array}{c} \mathcal{H} = \mathcal{F}(\mathcal{H}) \\ \bigcup \\ \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in \mathbf{Q}} \mathcal{H}_{\mathbf{k}} \end{array} \right] \\ \rho_{\mathcal{H}} = \sum_{\mathbf{k} \in \mathbf{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}} \rangle \langle \Psi_{N\mathbf{k}}^{\text{sv}}| & \in \mathcal{D} \left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\mathbf{k}} = L^2(\text{UC}) \otimes \mathbb{C}^2 = \left[\left(\bigoplus_{\alpha=1}^r L^2([0, R_{\alpha}], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \right) \oplus \left(L^2(\mathbb{R}) \otimes \mathbb{C}^2 \right) \right] \\ \bigcup \\ \rho_{N\mathbf{k}}^{\alpha} \end{array} \right] \\ \rho_{N\mathbf{k}} = |u_{N\mathbf{k}}^{\text{sv}} \rangle \langle u_{N\mathbf{k}}^{\text{sv}}| & \\ \rho_{N\mathbf{k}}^{\alpha} & \in \mathcal{D} \left[\begin{array}{c} \mathcal{H}_{\alpha} = L^2([0, R_{\alpha}], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \end{array} \right] \end{array}$$

$$\text{"Observable Quantity"} = \langle X^{2Q} \rangle = \text{Tr}_{\mathcal{F}} \left(\rho_{\mathcal{F}} X^{2Q} \right) = \text{Tr}_{\mathcal{H}} \left(\rho_{\mathcal{H}} X \right) = \dots$$

$$\begin{aligned} \dots &= N_p \sum_{\mathbf{k} \in \mathbf{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \text{Tr}_{\mathcal{H}_{\mathbf{k}}} \left(\rho_{N\mathbf{k}} X \right) \\ &= N_p \sum_{\mathbf{k} \in \mathbf{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \text{Tr}_{\mathcal{H}_{\alpha}} \left(\rho_{N\mathbf{k}}^{\alpha} X \right) \\ &\approx N_p \sum_{\mathbf{k} \in \mathbf{Q}} \sum_{N=1}^{\text{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \text{Tr}_{\mathcal{H}_{\alpha LS}} \left(\rho_{N\mathbf{k}}^{\alpha} X \right) \end{aligned}$$

$$\begin{array}{ll}
\rho_{\mathcal{F}} = \frac{1}{Z} \mathrm{e}^{-\beta(H^{2\mathrm{Q}} - \mu N^{2\mathrm{Q}})} & \in \mathcal{D}\left[\begin{array}{c} \mathcal{H} = \mathcal{F}(\mathcal{H}) \\ \bigcup \end{array}\right] \\
\rho_{\mathcal{H}} = \sum_{\mathbf{k} \in \mathcal{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\mathrm{sv}} \rangle \langle \Psi_{N\mathbf{k}}^{\mathrm{sv}}| & \in \mathcal{D}\left[\begin{array}{c} \mathcal{H} \equiv L^2(\mathrm{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in \mathcal{Q}} \mathcal{H}_{\mathbf{k}} \\ \bigcup \end{array}\right] \\
\rho_{N\mathbf{k}} = |u_{N\mathbf{k}}^{\mathrm{sv}} \rangle \langle u_{N\mathbf{k}}^{\mathrm{sv}}| & \in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\mathbf{k}} = L^2(\mathrm{UC}) \otimes \mathbb{C}^2 = \left[\left(\bigoplus_{\alpha=1}^r L^2([0, R_{\alpha}], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right) \oplus \left(L^2(\mathbb{R}) \otimes \mathbb{C}^2\right) \right] \\ \bigcup \end{array}\right] \\
\rho_{N\mathbf{k}}^{\alpha} & \in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\alpha} = L^2([0, R_{\alpha}], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \end{array}\right] \\
\rho_{\alpha LS}^{N\mathbf{k}} & \in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\alpha LS} = L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 = \bigoplus_{l=0}^{\infty} \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^l \bigcup_{\sigma \in \{\uparrow, \downarrow\}} \{Y_{lm} \otimes \chi_{\sigma} \equiv |lm\sigma\rangle\} \end{array}\right]
\end{array}$$

$$\text{"Observable Quantity"} = \langle X^{2\mathrm{Q}} \rangle = \mathrm{Tr}_{\mathcal{F}}\left(\rho_{\mathcal{F}} X^{2\mathrm{Q}}\right) = \mathrm{Tr}_{\mathcal{H}}\left(\rho_{\mathcal{H}} X\right) = \dots$$

$$\begin{aligned}
\dots &= N_p \sum_{\mathbf{k} \in \mathcal{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \mathrm{Tr}_{\mathcal{H}_{\mathbf{k}}} \left(\rho_{N\mathbf{k}} X\right) \\
&= N_p \sum_{\mathbf{k} \in \mathcal{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \mathrm{Tr}_{\mathcal{H}_{\alpha}} \left(\rho_{N\mathbf{k}}^{\alpha} X\right) \\
&\approx N_p \sum_{\mathbf{k} \in \mathcal{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \mathrm{Tr}_{\mathcal{H}_{\alpha LS}} \left(\rho_{\alpha LS}^{N\mathbf{k}} X\right)
\end{aligned}$$



$$\rho_{\mathcal{F}}=\frac{1}{Z}\mathrm{e}^{-\beta(H^{2Q}-\mu N^{2Q})}$$

$$\in \mathcal{D}\left[\begin{array}{c} \mathcal{H}=\mathcal{F}(H) \\ \bigcup \end{array}\right]$$

$$\rho_{\mathcal{H}}=\sum_{\mathbf{k}\in Q}\sum_{N=1}^{\text{nstsv}}\text{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)|\Psi_{N\mathbf{k}}^{\text{sv}}\rangle\langle\Psi_{N\mathbf{k}}^{\text{sv}}|$$

$$\in \mathcal{D}\left[\begin{array}{c} \mathcal{H}\equiv L^2(\text{Crystal})\otimes \mathbb{C}^2\simeq \bigoplus_{\mathbf{k}\in Q}\mathcal{H}_{\mathbf{k}} \\ \bigcup \end{array}\right]$$

$$\rho_{N\mathbf{k}}=|u_{N\mathbf{k}}^{\text{sv}}\rangle\langle u_{N\mathbf{k}}^{\text{sv}}|$$

$$\in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\mathbf{k}}=L^2(\text{UC})\otimes \mathbb{C}^2=\left[\left(\bigoplus_{\alpha=1}^rL^2([0,R_{\alpha}],r^2\mathrm{d}r)\otimes L^2(S^2,\mathrm{d}\Omega)\otimes \mathbb{C}^2\right)\oplus \left(L^2(\mathbb{IR})\otimes \mathbb{C}^2\right)\right] \\ \bigcup \end{array}\right]$$

$$\rho_{N\mathbf{k}}^{\,\,\alpha}$$

$$\in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\alpha}=L^2([0,R_{\alpha}],r^2\mathrm{d}r)\otimes L^2(S^2,\mathrm{d}\Omega)\otimes \mathbb{C}^2 \end{array}\right]$$

$$\rho_{\alpha LS}^{N\mathbf{k}}$$

$$\in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\alpha LS}=L^2(S^2,\mathrm{d}\Omega)\otimes \mathbb{C}^2=\bigoplus_{l=0}^{\infty}\mathrm{span}_{\mathbb{C}}\bigcup_{m=-l}^l\bigcup_{\sigma\in\{\uparrow,\downarrow\}}\left\{Y_{lm}\otimes\chi_{\sigma}\equiv|lm\sigma\rangle\right\} \end{array}\right]$$

$$\rho_{\alpha L(l)S}^{N\mathbf{k}}$$

$$\in \mathcal{D}\left[\begin{array}{c} \bigcup \\ \mathcal{H}_{\alpha L(l)S}=\mathrm{span}_{\mathbb{C}}\bigcup_{m=-l}^l\bigcup_{\sigma\in\{\uparrow,\downarrow\}}\left\{Y_{lm}\otimes\chi_{\sigma}\equiv|lm\sigma\rangle\right\} \end{array}\right]$$

$$\text{"Observable Quantity"}=\langle X^{2Q}\rangle=\text{Tr}_{\mathcal{F}}\left(\rho_{\mathcal{F}}X^{2Q}\right)=\text{Tr}_H\left(\rho_HX\right)=...$$

$$\begin{aligned} & \dots=N_p\sum_{\mathbf{k}\in Q}\sum_{N=1}^{\text{nstsv}}\text{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)\text{Tr}_{H_{\mathbf{k}}}\left(\rho_{N\mathbf{k}}X\right)\\ &=N_p\sum_{\mathbf{k}\in Q}\sum_{N=1}^{\text{nstsv}}\text{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^r\text{Tr}_{H_{\alpha}}\left(\rho_{N\mathbf{k}}^{\alpha}X\right)\\ &\approx N_p\sum_{\mathbf{k}\in Q}\sum_{N=1}^{\text{nstsv}}\text{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^r\text{Tr}_{H_{\alpha LS}}\left(\rho_{\alpha LS}^{N\mathbf{k}}X\right)\\ &=N_p\sum_{\mathbf{k}\in Q}\sum_{N=1}^{\text{nstsv}}\text{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^r\sum_{l=0}^{\infty}\text{Tr}_{H_{\alpha L(l)S}}\left(\rho_{\alpha L(l)S}^{N\mathbf{k}}X\right) \end{aligned}$$

$$\rho_{\mathcal{F}} = \frac{1}{Z} e^{-\beta(H^{2Q} - \mu N^{2Q})}$$

$$\in \mathcal{D} \left[\begin{array}{c} \mathcal{H} = \mathcal{F}(H) \end{array} \right]$$

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$$\rho_H = \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}} \rangle \langle \Psi_{N\mathbf{k}}^{\text{sv}}|$$

$$\in \mathcal{D} \left[\begin{array}{c} \mathcal{H} \equiv L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in Q} \mathcal{H}_{\mathbf{k}} \end{array} \right]$$

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$$\bigcup$$

$$\rho_{N\mathbf{k}}^{\alpha}$$

$$\in \mathcal{D} \left[\begin{array}{c} \mathcal{H}_{\alpha} = L^2([0, R_{\alpha}], r^2 dr) \otimes L^2(S^2, d\Omega) \otimes \mathbb{C}^2 \end{array} \right]$$

$$\bigcup$$

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$$\in \mathcal{D} \left[\begin{array}{c} \mathcal{H}_{\alpha LS} = L^2(S^2, d\Omega) \otimes \mathbb{C}^2 = \bigoplus_{l=0}^{\infty} \text{span}_{\mathbb{C}} \bigcup_{m=-l}^l \bigcup_{\sigma \in \{\uparrow, \downarrow\}} \{Y_{lm} \otimes \chi_{\sigma} \equiv |lm\sigma\rangle\} \end{array} \right]$$

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$$\in \mathcal{D} \left[\begin{array}{c} \mathcal{H}_{\alpha L(l)S} = \text{span}_{\mathbb{C}} \bigcup_{m=-l}^l \bigcup_{\sigma \in \{\uparrow, \downarrow\}} \{Y_{lm} \otimes \chi_{\sigma} \equiv |lm\sigma\rangle\} \end{array} \right]$$

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for example ...

- the spin-operator $X \equiv S_i$ with $i \in \{x, y, z\}$,
- the angular-momentum operator $X \equiv L_i$ with $i \in \{x, y, z\}$,
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$$\bigcup$$

$$\rho_{N\mathbf{k}}^{\alpha}$$

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$$\rho_{\alpha L S}^{N\mathbf{k}}$$

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$$\text{"Observable Quantity"} = \langle X^{2Q} \rangle = \text{Tr}_{\mathcal{F}} (\rho_{\mathcal{F}} X^{2Q}) = \text{Tr}_H (\rho_H X) = \dots$$

$$\dots = N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \text{Tr}_{H_{\mathbf{k}}} (\rho_{N\mathbf{k}} X)$$

$$= N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \text{Tr}_{H_{\alpha}} (\rho_{N\mathbf{k}}^{\alpha} X)$$

$$\approx N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \text{Tr}_{H_{\alpha L S}} (\rho_{\alpha L S}^{N\mathbf{k}} X)$$

$$= N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \sum_{l=0}^{\infty} \text{Tr}_{H_{\alpha L(l)S}} (\rho_{\alpha L(l)S}^{N\mathbf{k}} X)$$

$$\dim \mathcal{H}_{\{\alpha L(l)S\}}$$

$$=$$

$$2 * [l] = 2 * (2 * l + 1)$$

for example ...

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Notation 7.1 For the discussion of tensor-moments, we consider a density-matrix ρ that has the dimensions $2[l] \times 2[l]$ with matrix elements denoted as

$$\rho \equiv \rho_{ms,m's'} \in \text{Herm}_{2[l] \times 2[l]}(\mathbb{C}) \tag{7.1}$$

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We will now introduce

triple upper-index tensor-moments $w_t^{kpr} \in \mathbb{C}$ with $k \in \{0, 1, 2, \dots, 2l\}$, $p \in \{0, 1\}$, $r \in \{|k-p|, \dots, |k+p|\}$
and $t \in \{-r, \dots, +r\}$

$$w_t^{kpr} = \text{Tr} \left(\Gamma_t^{kpr} \rho \right)$$

with *tensor-matrices*⁵

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$$\Gamma_{t,ab}^{kpr} = N_{kprls}^{-1} (-1)^{l-m_a+s-s_a} \sum_{\substack{x \in \{-k, \dots, +k\} \\ y \in \{-p, \dots, +p\}}} (-1)^{+k-x+p-y} \begin{pmatrix} l & k & l \\ -m_a & x & m_b \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_a & y & s_b \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix},$$

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$$[l] = 2 * l + 1$$

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$$w_t^{kpr} = \text{Tr} \left(\Gamma_t^{kpr} \rho \right) \iff \rho_{ab} = \sum_{kp rt} \left(\Lambda_t^{kpr} \right)_{ab} \cdot w_t^{kpr}$$

with *tensor-matrices*⁵

$$\Gamma_{t,ab}^{kpr} = N_{kprls}^{-1} (-1)^{l-m_a+s-s_a} \sum_{\substack{x \in \{-k, \dots, +k\} \\ y \in \{-p, \dots, +p\}}} (-1)^{+k-x+p-y} \begin{pmatrix} l & k & l \\ -m_a & x & m_b \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_a & y & s_b \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix},$$

$$\Lambda_{t,ab}^{kpr} = N_{kprls} [kpr] (-1)^{m_b-l+s_b-s} \sum_{\substack{x \in \{-k, \dots, +k\} \\ y \in \{-p, \dots, +p\}}} (-1)^{x-k+y-p} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}$$

Theorem 8.2 We obtain

$$\mathrm{Tr} \left(\rho \mathbb{1} \right) \equiv \mathrm{Tr} \, \rho = w_0^{000} \tag{8.4}$$

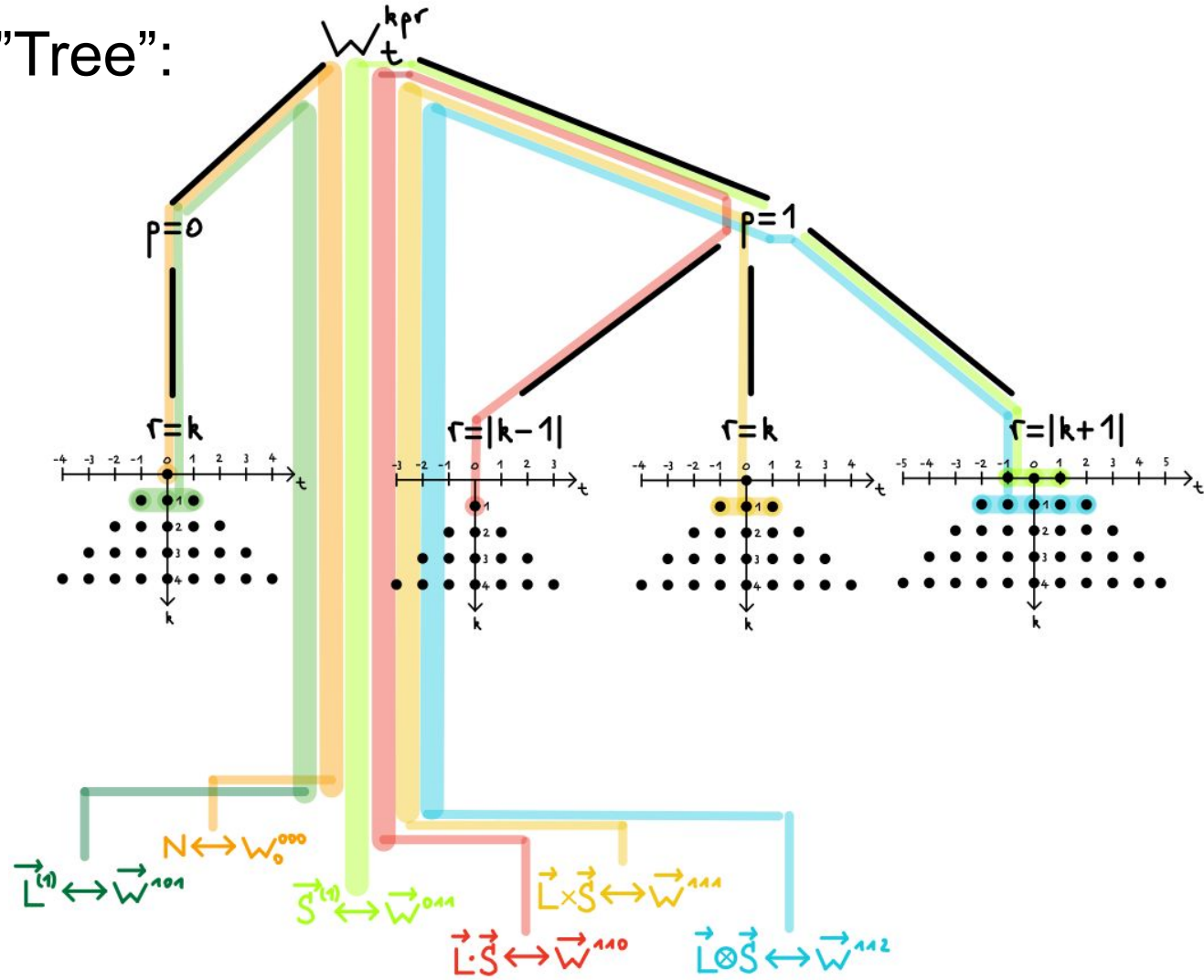
$$\mathrm{Tr} \left[\rho \mathcal{S}_q^{(1)} \right] = -s w_q^{011} \tag{8.6}$$

$$\mathrm{Tr} \left[\rho L_q^{(1)} \right] = -l w_q^{101} \tag{8.7}$$

$$\mathrm{Tr} \left[\rho \left(\mathbf{L} \cdot \mathbf{S} \right) \right] = l \cdot s \cdot w_0^{110} \tag{8.8}$$

$$\text{"Observable Quantity"} = \langle X^{2Q} \rangle = \dots \approx N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \sum_{l=0}^{\infty} \mathrm{Tr}_{\mathcal{H}_{\alpha L(l)S}} \left(\rho_{\alpha L(l)S}^{N\mathbf{k}} X \right) \tag{8.1}$$

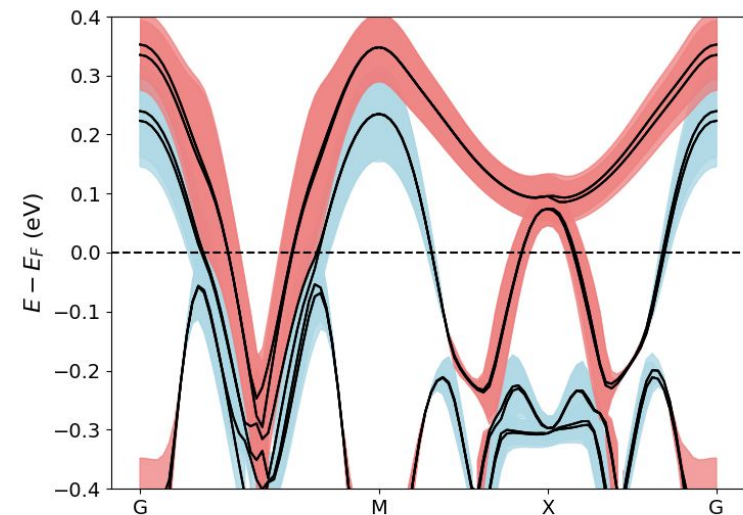
Tensor-Moment-“Tree”:



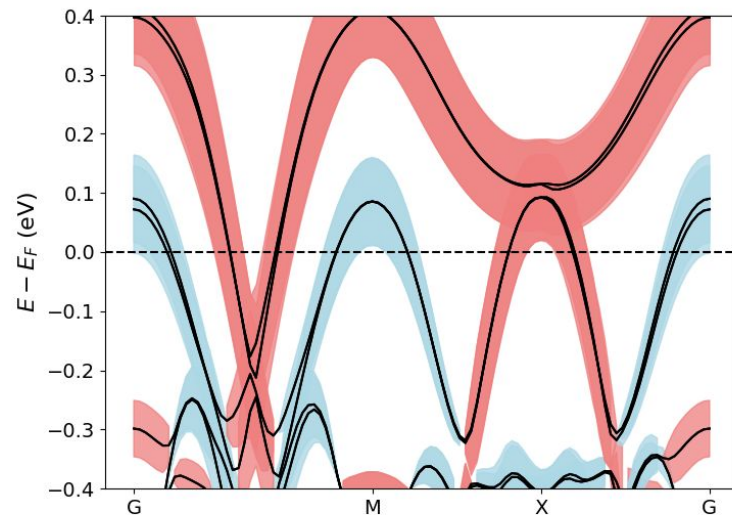
Featuring:

- Bandstructure with Tensor-Moment 110 for Rh-atoms, i.e. $w_{N\mathbf{k}\alpha l}^{110}$ as linewidth with $l = 2$ and $\alpha \longleftrightarrow \text{Rh}$
- blue / red LW = positive / negative sign

$U = 0.0 \text{ eV}$

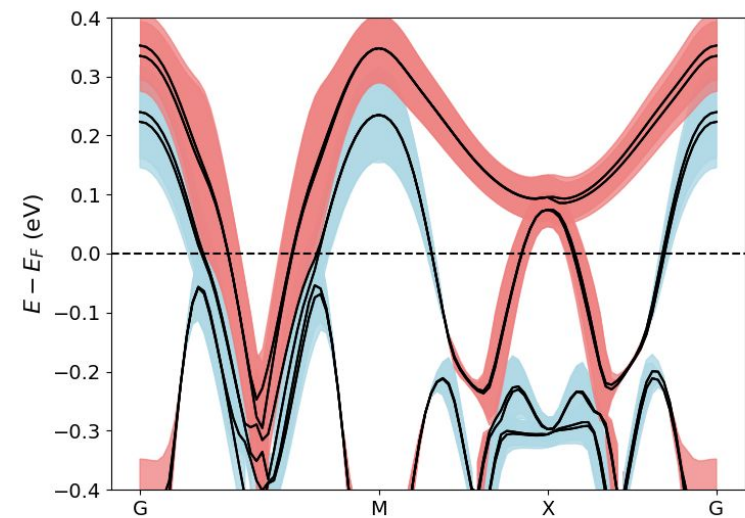


$U = 3.6 \text{ eV}$

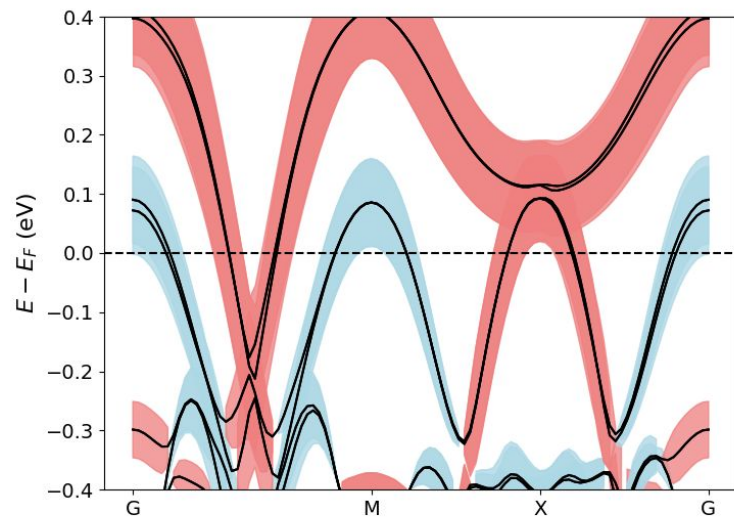


“Behind the Scenes” ...

$U = 0.0$ eV



$U = 3.6$ eV

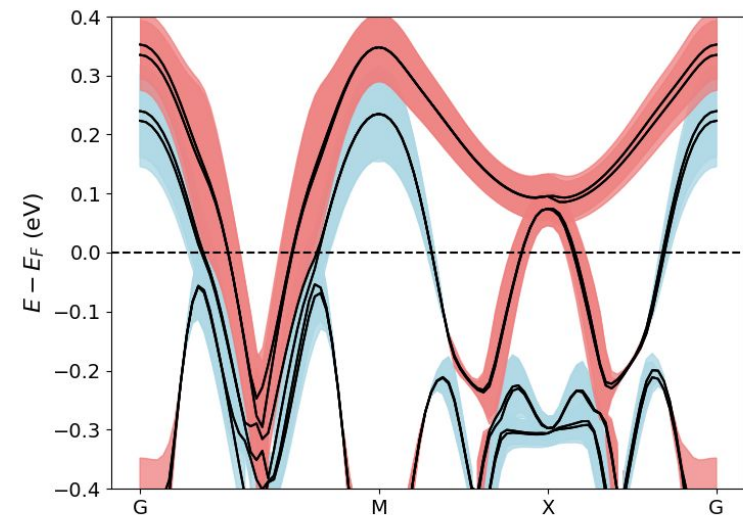


Featuring:

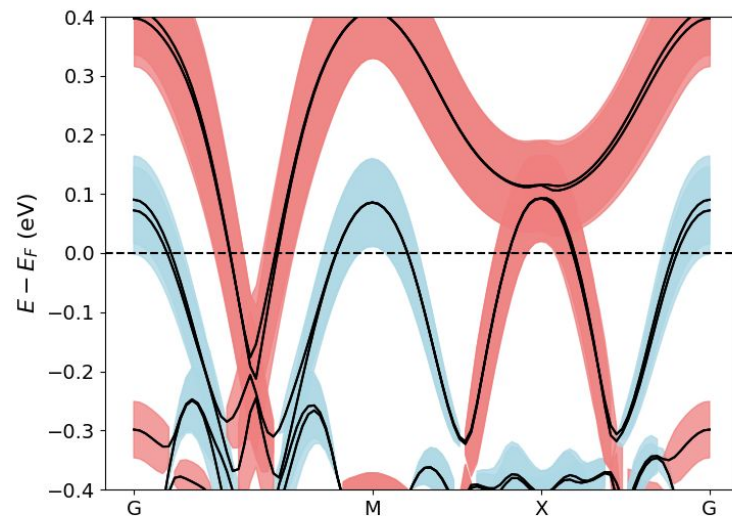
- Bandstructure with Tensor-Moment 110 for Rh-atoms, i.e. $w_{Nk\alpha l}^{110}$ as linewidth with $l = 2$ and $\alpha \longleftrightarrow \text{Rh}$
- blue / red LW = positive / negative sign

$$\langle (\mathbf{L} \cdot \mathbf{S})^{2Q} \rangle \approx N_p \sum_{\mathbf{k} \in Q} \sum_{N=1}^{\text{nstsv}} F_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^r \sum_{l=0}^{\infty} l \cdot s \cdot w_{Nk\alpha l}^{110}$$

$U = 0.0 \text{ eV}$



$U = 3.6 \text{ eV}$

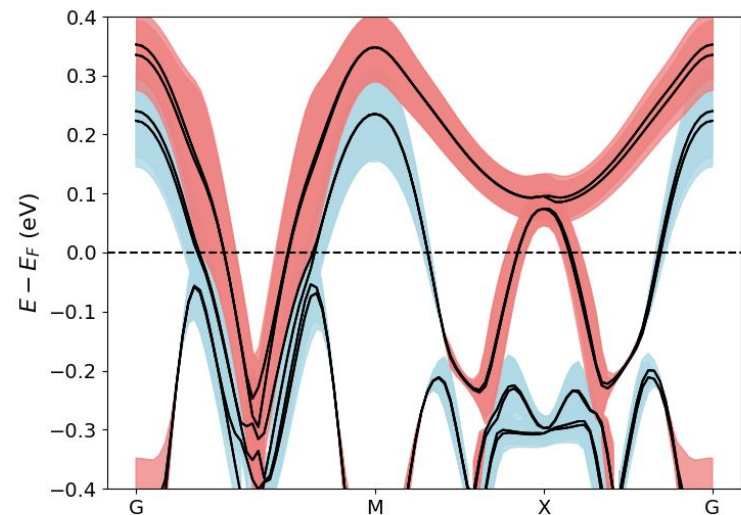


Featuring:

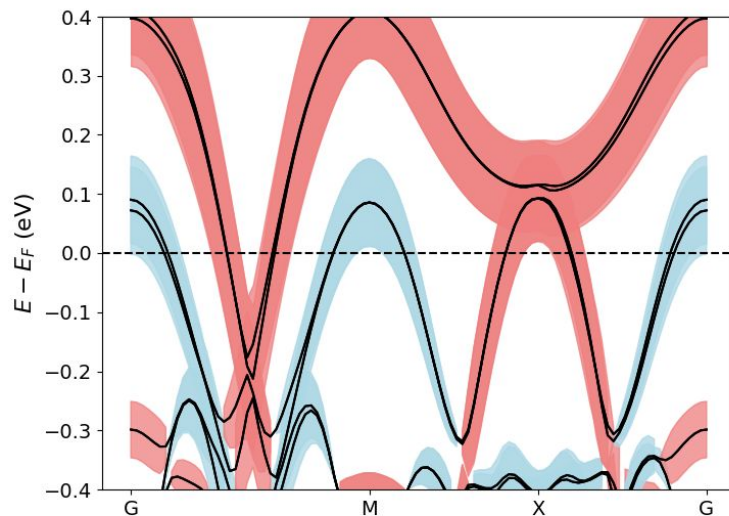
- Bandstructure with Tensor-Moment 110 for Rh-atoms as linewidth
- blue / red LW = positive / negative sign

Interpretation: The spin-orbit-interaction of the system increases, when U increases!

$U = 0.0$ eV



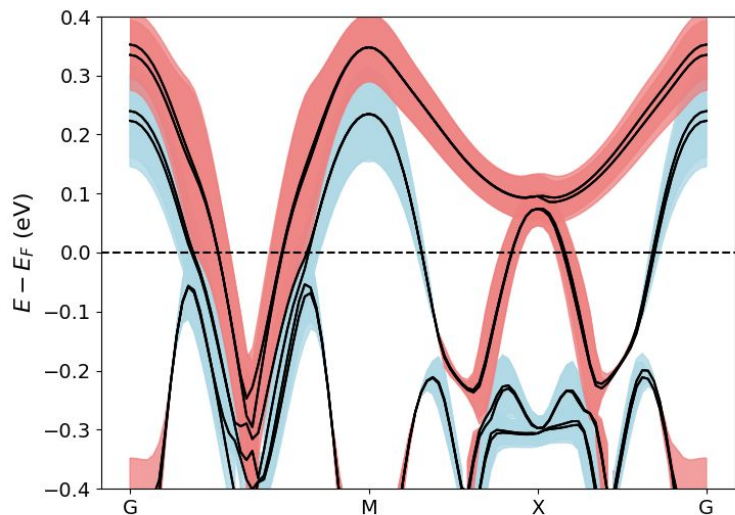
$U = 3.6$ eV



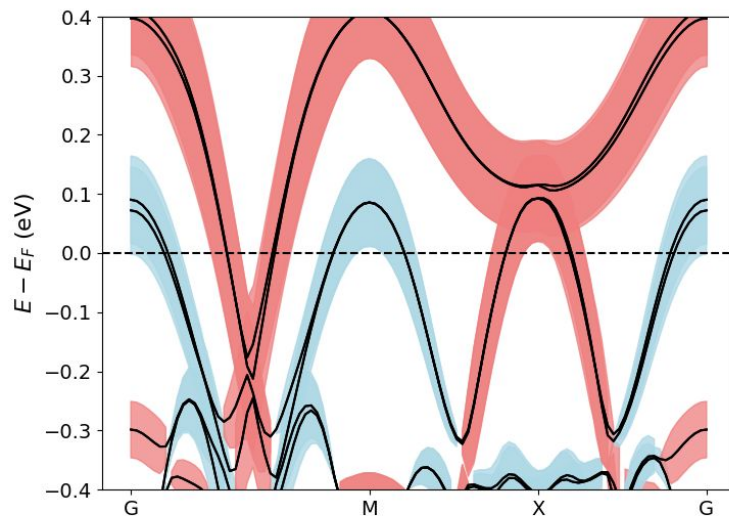
Hypothesis:

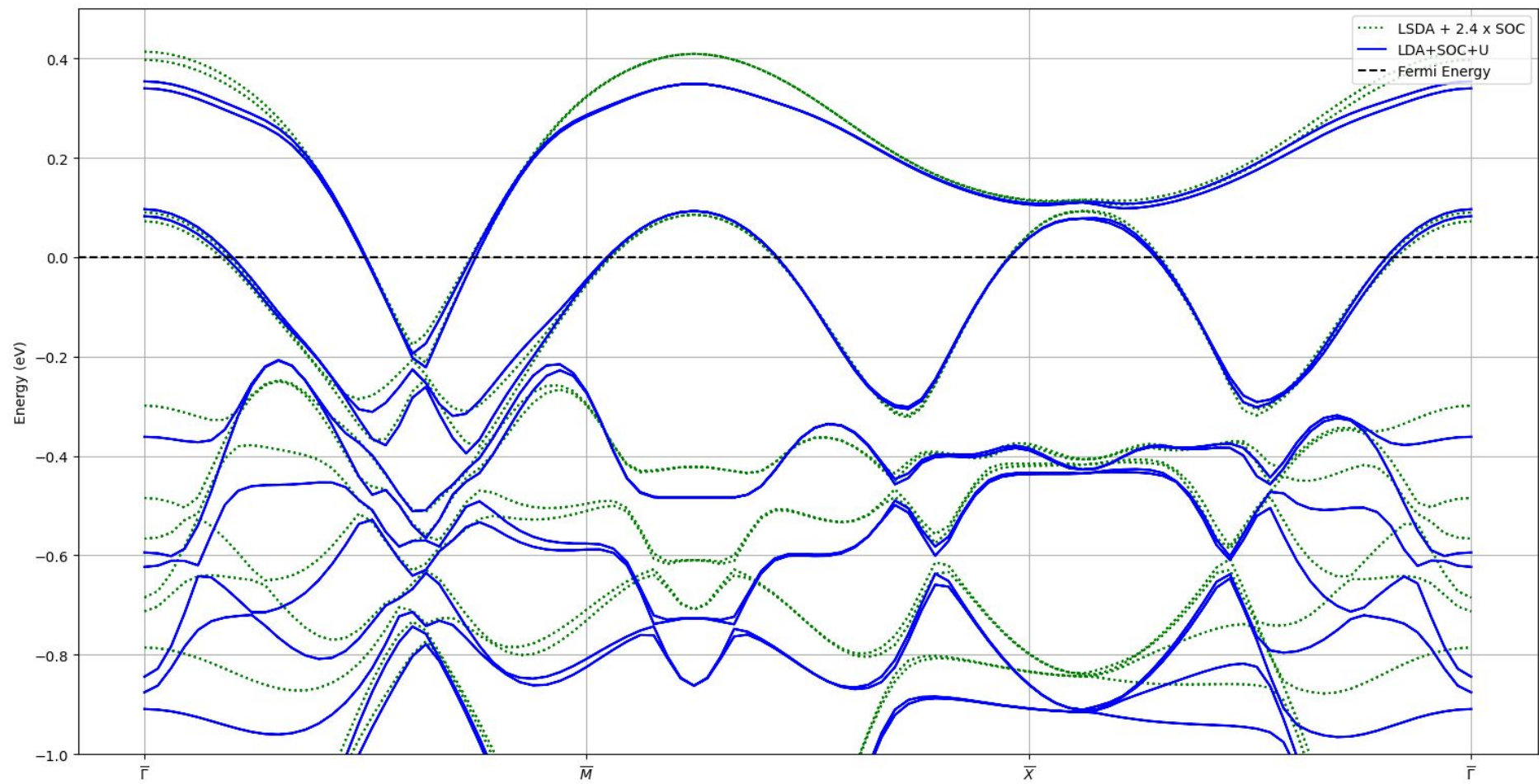
- The shift in energy bands corresponds to an increase in spin-orbit-interactions!

$U = 0.0 \text{ eV}$



$U = 3.6 \text{ eV}$





5 / 5 : Weyl-Points

Fact:

- In order to find Weyl-Points, one always has to do the same three steps!
- **Step 1:**
Determining Fermi-Energy, Bandstructure and Projected DOS
- **Step 2:**
Finding a good Wannierization with Wannier90 (based on Step 1)
- **Step 3:**
Using the Wannier-Tools software to find the crossings on fine grids (based on Step 2).

[Currently less than a “draft”, but much more than just “begun”]



Kalhöfer-Technologies

Welcome to Kalhöfer-Technologies.

Choose a service:

ELK

ELK Demonstration

GRIT

GRIT Demonstration

AI

AI Demonstration



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My Material Calculations

Available Materials

My Material Calculations

Create a new Material Calculation

WTe2

| ?

| Abhirup the Magician



TaAs

| 141md

| TBAdded from ICSD



Li

| 1m-3m

Analytical Chemistry (Washingt...)



Fact:

- In order to find Weyl-Points, one always has to do the same three steps!
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My Profile

My Input Preferences

My Material Calculations

Available Materials

Support

Logout

TaAs | I41md | TBAdded from ICSD

Step 1

Step 2

Step 3

Not interested in this calculation

Edit Step 1 Input Parameters

Download Step 1 Inputs

Computing Machine

Calculations

☐ PC ☒ Dardel

☒ elk-scf ☒ elk-dos ☒ elk-bands

Crystal Information

Lattice Vectors

avec

1.0000000000	0.0000000000	0.0000000000
0.0000000000	1.0000000000	0.0000000000
0.5000000000	0.5000000000	1.6956648240

Basis Vectors

atoms

2 : nspecies

'Ta.in' : spfname

2

0.0000000000000000	0.0000000000000000	0.0000000000000000	0.00000000	0.00000000	0.00000000
0.7500000000000000	0.2500000000000000	0.5000000000000000	0.00000000	0.00000000	0.00000000

'As.in' : spfname

Convergence settings

ngridk

swidth

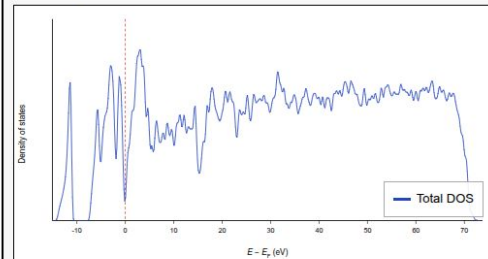
Fermi energy

Energy (eV)

11.401

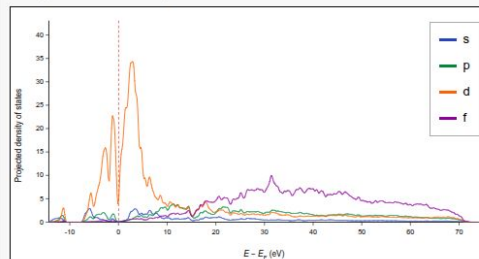
EC-3-3

Density of states



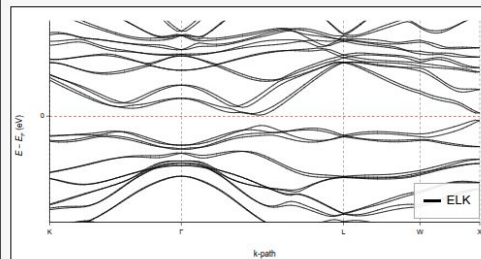
Reset zoom

Projected density of states — S01



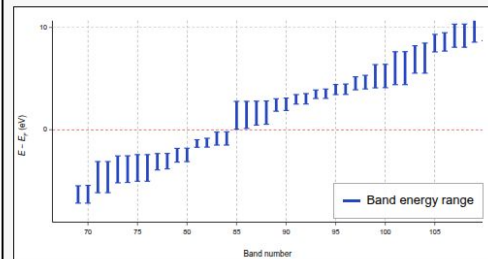
Reset zoom

Band structure



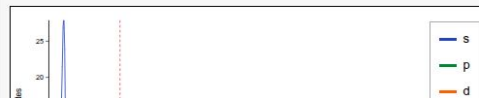
Reset zoom

Band widths



Reset zoom

Projected density of states — S02



Electronic structure dashboard

Convergence settings

ELK convergence settings

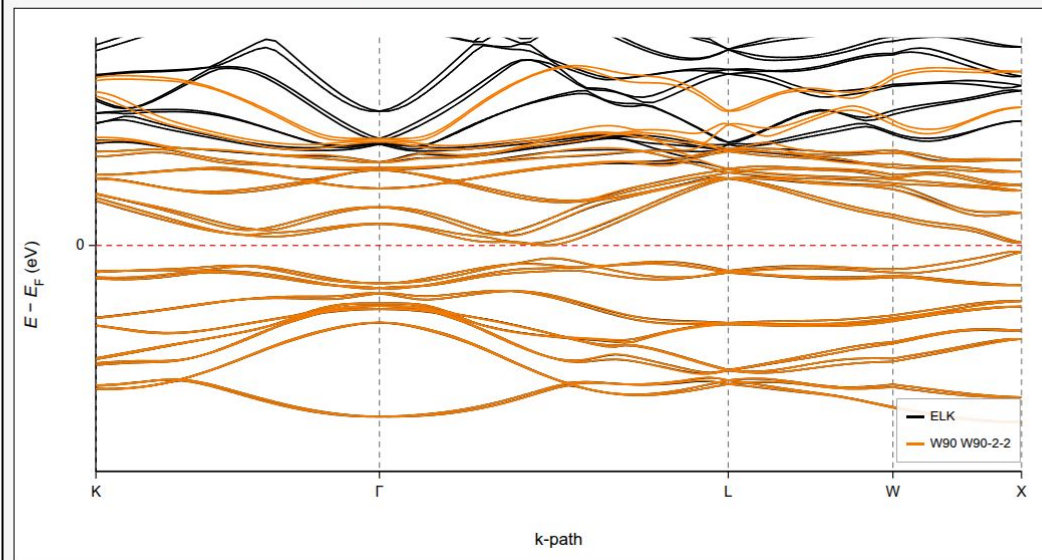
ngridk
16 16 16

W90 convergence settings

num_iter
2000

dis_num_iter
2000

ELK/Wannier90 band-structure comparison



Wannier-function centres and spreads

W90-2-2

WF	x	y	z	Spread
9	0.099936	0.036893	0.025839	8.585433
11	0.114421	0.039836	-0.029304	11.944416
4	0.039698	-0.140472	-0.022769	10.443179

ELK/Wannier90/WannierTools band comparison

Convergence settings

ELK convergence settings

No convergence parameters configured.

W90 convergence settings

No W90 convergence parameters configured.

WannierTools convergence settings

NumOccupied

16

Nki

15

Gap_threshold

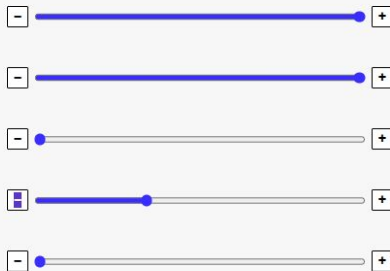
0.001

delta

0.002

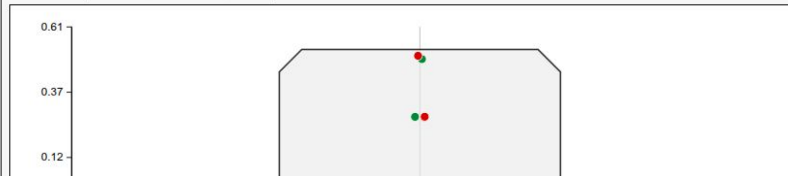
Nki_spherical

41

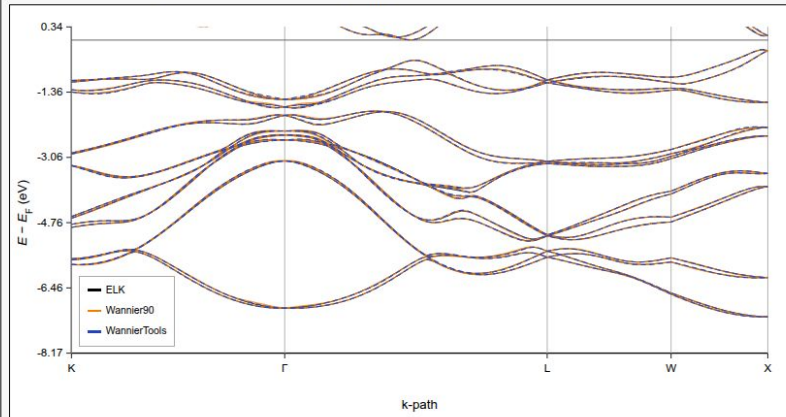


Weyl points and chiralities

N = 26 (+1: 13, -1: 13, 0: 0)



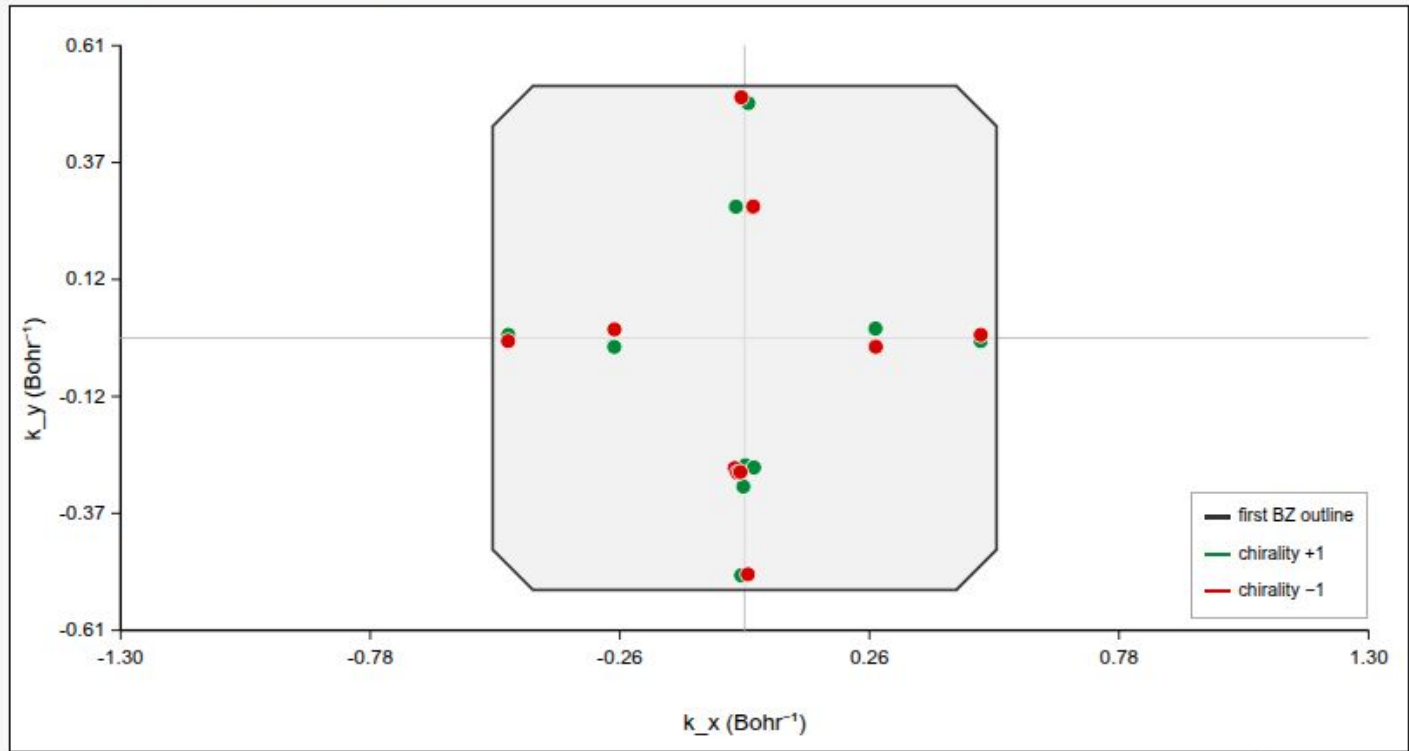
ELK/Wannier90/WannierTools band-structure comparison



Reset zoom

Weyl points and chiralities

N = 26 (+1: 13, -1: 13, 0: 0)



Reset zoom

Hide chirality 0 points

WTe₂ (in Progress ...)

Afterwards: Application to “less studied materials”

Which Materials require Hubbard interactions and possess Weyl-points?

Yes. If by “**require Hubbard interactions**” you mean materials for which plain semilocal DFT is inadequate or questionable and a local Coulomb term U —via DFT+ U , Hubbard models, or preferably DMFT—is important for obtaining the relevant magnetic/electronic structure, then there is a rather interesting overlap with **magnetic Weyl materials**.

The most compelling examples are not the canonical TaAs-family Weyl semimetals; those are comparatively weakly correlated. Instead, I would look at **pyrochlore iridates**, **correlated magnetic oxides**, **kagome magnets**, and **rare-earth Weyl semimetals**.

Material / family	Correlated orbitals	Typical treatment	Weyl character	How strongly I would recommend it
$R_2\text{Ir}_2\text{O}_7$, especially $\text{Y}_2\text{Ir}_2\text{O}_7$, $\text{Pr}_2\text{Ir}_2\text{O}_7$	Ir $5d$	Hubbard U +SOC / DFT+ U / DMFT	Magnetic Weyl nodes near correlation-driven MIT	Excellent
$\text{Co}_3\text{Sn}_2\text{S}_2$	Co $3d$	DFT+ U , DFT+DMFT	Experimentally established magnetic Weyl semimetal	Excellent
SrRuO_3	Ru $4d$	DFT+ U , DMFT	Multiple Weyl nodes near E_F	Excellent
HgCr_2Se_4	Cr $3d$	LDA+ U +SOC	Predicted double-Weyl nodes	Very interesting theoretically
Mn_3Sn	Mn $3d$	DFT+ U sometimes used	Magnetic Weyl semimetal	Good , but U is less essential
CeAlGe / CeAlSi	Ce $4f$	DFT+ U 	Noncentrosymmetric magnetic Weyl	Excellent for f-electron physics

Acknowledgements

Acknowledgements

“Trio Infernale”:

- **Jonas** (GREEN-FUNCTIONS (!!!))
- **Lars** (DFT, ELK, Datathief)
- **Oscar** (DFT, ELK)

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- Lucas (Manuscript Proof-Reading, Advice)
- Leonard (Eliashberg-Theory Book and Residue-Thm)
- Mahroo (Latex-templates and Green-Fcts)
- Manuel (Advice on Magnetism)
- Matthias (General Advice)
- Moa (Moral Support)
- Maxime (Feynman-Diagrams)
- Nastaran (Moral Support and Fortran)
- Paul (Phono3py)
- Patrik (Chemical Potential (!) and Advice)
- Peter (Convos about EI-Ph-Interactions)
- Prathibha (Moral Support and SQL)
- Quentin (expectation-value of position-operators)
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- Sanja (Moral Support and Advice)
- Sanaz (Moral Support)
- Sofiya (QFT-related questions)
- Soheil (Wannier90, Course-Help)
- Susanne (General Advice)
- Sushant (Advice on symmetry-breaking)
- Vladislav (DFT, MIDNIGHT COMMANDER (!!!))
- ... (?)

Friends and Colleagues (alphabetically ordered):

- Abhirup (Weyl-Points, Spintextures, Wannierizations)
- Alena (DFT)
- Andreas T. (Linux and workshop)
- Angel (Moral Support)
- Annica (Manuscript Proof-Reading, Advice)
- Arnob (Submissions, Latex, Advice)
- Atefe (Moral Support and AI)
- Aksel (Automatizing Installations)
- Biplab (Course and Avice)
- Chin (DFT, QE, GRIT (!!!), Parallelizations)
- Enrico (Grassmann-identities)
- Fabian (Eliashberg-Theory)
- Family (Moral Support)
- Federica (Moral Support and Moving)
- Felix (DFT+U)
- Florian 1 (Linux and other computer-related issues)
- Florian 2 (expectation-value of position-operator)
- Francesco (the one before the last one, Green-Fcts.)
- Georgia (DFT+U)
- Johan (DARDEL (!!!))
- Jakub (ELK-scripts and Dardel kickstart)

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... and from (almost) everyone: Lots and lots and lots of **encouragement**. <3

Thank you (for the music)!



Questions / Suggestions / Comments?